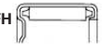
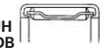
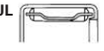
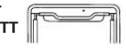
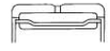
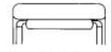
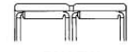
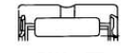
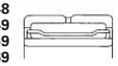
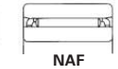
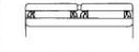
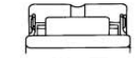
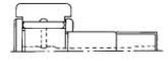
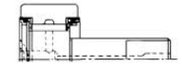
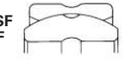
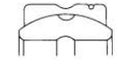
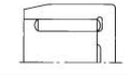
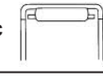
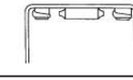
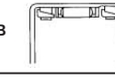
MOTION & CONTROL™

NSK

Needle Roller Bearings



NSK NEEDLE BEARING

TECHNICAL INFORMATION		A6～	
Cage & Needle Roller Assemblies	FWJ FWF WJ  FWF---W  FBN WJC  FBN---W 	B4～	
Drawn Cup Needle Roller Bearings	F, FH NFH B, BH  MF, MFH M, MH  FJ, FJL, FJH J, JH, DB  MFJ, MFJL MFJH MJ, MJH  Y YH DD  MFY  FJT, FJTT FJLT, FJLTT FJHTT  MFJT MFJLT  FJP JP 	B36～	
Solid Needle Roller Bearings	RNA48 RNA49 RNA59 RNA69 HJ  RLM  RNAF  RNAFW  RNA---TT  NA48 NA49 NA59 NA69 HJ+IR  LM  NAF  NAFW  NA---TT 	B88～	
Inner Rings for Needle Roller Bearings	FIR IR  IR 	B128～	
Thrust Bearings	FNTA NTA  FB  FG  FH  FTRA TRA FTRB TRB FTRC TRC FTRD TRD  FTRE TRE TRF 	B162～	
Cam Followers	FCR FCJ CR  FCRS FCJS CRS 	B176～	
Roller Followers	FYCR FYCJ YCR  FYCRS FYCJS YCRS 	B190～	
Needle Rollers	A F P T C M	B198～	
Spherical Plain Bearings	FSF SF  FSBB SBB 	B208～	
Needle Roller Bearings for Universal Joints	ZY  NSA 	B220～	
Drawn Cup Roller Clutches	FC  RC  FCB  RCB 	B228～	
Oil Seals for Needle Roller Bearings	VC  KC 	B238～	
Example Figures and Appendices		C1～	

Introduction

Newly Revised & Updated NSK Needle Roller Bearing Catalog

Welcome to the latest edition of the NSK Needle Roller Bearing catalog (Catalog No. E1419c). Our new catalog has been revised and updated to better meet the needs of our customers.

NSK needle roller bearings enjoy a favorable reputation and are well accepted in the industrial machinery marketplace. NSK will continue to meet the future needs of industry by further developing automation-related products that are more compact, lightweight, consume less energy, and offer extended maintenance-free performance.

This newly revised NSK needle roller bearing catalog has been enhanced with a wealth of information that spans many years of amassing empirical data from many sources around the world.

We are convinced that this catalog will prove to be more helpful to you in selecting the optimum bearing for your application. Choose the optimal NSK needle roller bearing for your application from among the types and features offered in this catalog. Please feel free to contact us with any feedback, comments or suggestions you may have.

MOTION & CONTROL
NSK

Needle Roller Bearings

CAT.No.E1419c

CONTENTS

TECHNICAL INFORMATION Page

1. Bearing Types and Features ..... A 6
1.1 Cage & Needle Roller Assemblies .... A 6
1.2 Drawn Cup Needle Roller Bearings .. A 6
1.3 Solid Needle Roller Bearings ..... A 8
1.4 Thrust Bearings ..... A 8
1.5 Cam Followers and Roller Followers A 9
1.6 Needle Rollers ..... A 9
1.7 Spherical Plain Bearings A10
1.8 Needle Roller Bearings for Universal Joints A10
1.9 Drawn Cup Roller Clutches A11

2. Bearing Life and Basic Load Rating .. A12
2.1 Bearing Life A12
2.2 Basic Dynamic Load Rating A12
2.3 Life Equation A13
2.4 Static Load Rating A15
2.5 Maximum Permissible Load of Drawn Cup Needle Roller Bearings, P_{max} A16

3. Bearing Loads A17
3.1 Equivalent Load A17
3.2 Calculation of Bearing Loads A17
3.3 Average Load A18

4. Bearing Selection A20
4.1 Bearing Types A20
4.2 Machinery in which Bearings are Used and Projected Life A22
4.3 Limiting Speed A22
4.4 Internal Clearance of Rolling Bearings A23
4.5 Bearing Accuracy A25

5. Shaft and Housing Design A30
5.1 Fitting A30
5.2 Accuracy and Roughness A34
5.3 Materials and Heat Treatment of Raceway Surface A35
5.4 Abutment and Fillet Dimensions A35
6. Lubrication and Lubricating Methods .. A37
6.1 Friction and Lubrication A37
6.2 Lubricants A37
6.3 Lubricating Methods A41

7. Bearing Seals A43
7.1 Oil Seals A43
7.2 Oil Grooves A43
7.3 Flingers (Slingers) A44
7.4 Labyrinth Seals A44

8. Bearing Handling A45
8.1 Precautions for Proper Handling of Bearings A45
8.2 Mounting A45
8.3 Insertion and Removal Force A46
8.4 Operating Inspection A47
8.5 Dismounting A47
8.6 Cleaning A48

BEARING TABLES Page

Cage & Needle Roller Assemblies ..... B 4
Drawn Cup Needle Roller Bearings ..... B 36
Solid Needle Roller Bearings ..... B 88
Inner Rings for Needle Roller Bearings ... B128
Thrust Bearings B162
Cam Followers B176
Roller Followers B190
Needle Rollers B198
Spherical Plain Bearings B208
Needle Roller Bearings for Universal Joints .. B220
Drawn Cup Roller Clutches B228
Oil Seals for Needle Bearings B238

EXAMPLE FIGURES Page

- Motor Vehicles ..... C 2
- Construction and Cargo Handling Equipment ..... C 9
- Agricultural Machinery C19
- Machine Tools C20
- Business Machines C26
- Electric and Pneumatic Equipment C30
- Two-Stroke Engines C39
- Hydraulic Machines C41
- Others C44

APPENDICES

Appendix 1 Conversion Table from SI (International Units) System C48
Appendix 2 N-kgf Conversion Table C50
Appendix 3 kg-lb Conversion Table C51
Appendix 4 °C-°F Conversion Table C52
Appendix 5 Viscosity Conversion Table C53
Appendix 6 Inch-mm Conversion Table C54
Appendix 7 Hardness Conversion Table (Reference) ... C56
Appendix 8 Physical and Mechanical Properties of Materials C57
Appendix 9 Tolerances for Shaft Diameters C58
Appendix 10 Tolerances for Housing Bore Diameters ... C60
Appendix 11 Values of Standard Tolerance Grades IT .. C62
Appendix 12 Speed Factor f_n C64
Appendix 13 Fatigue Life Factor f_h and Fatigue Life $L \cdot L_h$ C65
Appendix 14 Boundary Dimensions of Radial Bearings ... C66
Appendix 15 Boundary Dimensions of Thrust Bearings ... C67

1. Bearing Types and Features

Needle roller bearings are a type of roller bearings and are classified as either radial or thrust depending on the direction of the load they support. Needle roller bearings include bearings whose rollers slightly exceed the size range of needle rollers as stipulated by ISO.

Needle roller bearings include drawn cup and solid radial bearings, as well as application-specific cam followers, roller followers. Thrust bearings include thrust needle bearings. Although not classified as a rolling bearing, there is also a spherical sliding bearing that supports radial and axial loads.

This catalog presents the various types of needle bearings and their features. For easy reference, the numbers in each section from **1.1** through **1.0** correspond to the numbers in **Figs. 1.1** through **1.9**.

1.1 Cage & Needle Roller Assemblies

1. The cage has a unique structure that facilitates handling by holding the rollers securely in place.

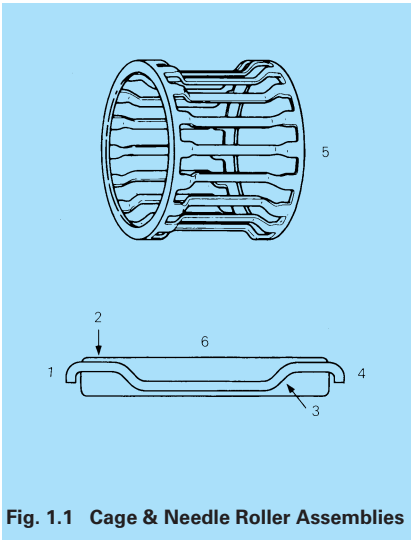


Fig. 1.1 Cage & Needle Roller Assemblies

2. The outside surface of the cage serves as a guide face and helps create favorable lubricating conditions.
3. The cage guides the rollers accurately to ensure that stable roller movement is maintained.
4. The single-piece cage has high strength and rigidity and facilitates the free circulation of lubricant.
5. High load capacity is possible due to the presence of many rollers of relatively long length.
6. To prevent stress concentrations on both ends of the rollers (edge loads), appropriate crowning is applied.

1.2 Drawn Cup Needle Roller Bearings

1. NSK drawn cup needle roller bearings are available in two types: caged and full complement. Drawn cup outer rings are made of carefully selected alloy steel sheet. They are surface-hardened after precise pressing, giving them a unique structure.
2. The outer ring is precisely pressed and is in tight line contact with the rollers. Optimum heat treatment gives the raceway surface uniform strength and the ability to carry heavy loads.
3. The low height of the bearings saves space so machines can be made more compact and lightweight.
4. The bent part of the drawn cup outer ring securely holds the rollers and/or cage and acts as a labyrinth seal that prevents both the entry of dust and dirt and lubricant leakage. At the same time, it enhances the strength of the outer ring end faces.
5. When the shaft has adequate hardness and is finished to proper dimensional accuracy, the inner ring can be omitted and space in the radial direction can be saved.
6. As securing in the axial direction is achieved by press fitting, a locating snap ring, collar, housing shoulder or the like is not required. This simplifies the housing structure and facilitates low-cost design.

7. Mounting is made easier because press fitting can be performed with an arbor press or the like.
8. Lubricating the bearing is easy and a lubrication hole can also be provided upon request.
9. If a closed type is used at the shaft end, savings can be made on the housing end cover.
10. The full complement roller bearing carries the load on many rollers and thus maximizes load capacity.
11. Among full complement roller bearings, bearings whose rollers are held in place by grease before assembly have the maximum load capacity because the rollers have the longest possible length.
12. The cage of the needle bearing accurately guides the rollers at the pitch diameter of the rollers. The surface of the cage is hardened to reduce frictional torque and increase rigidity and wear resistance.
13. The unique structure of the cage allows extensive space for storing lubricant, thus enabling smooth lubricant circulation and increasing grease life.

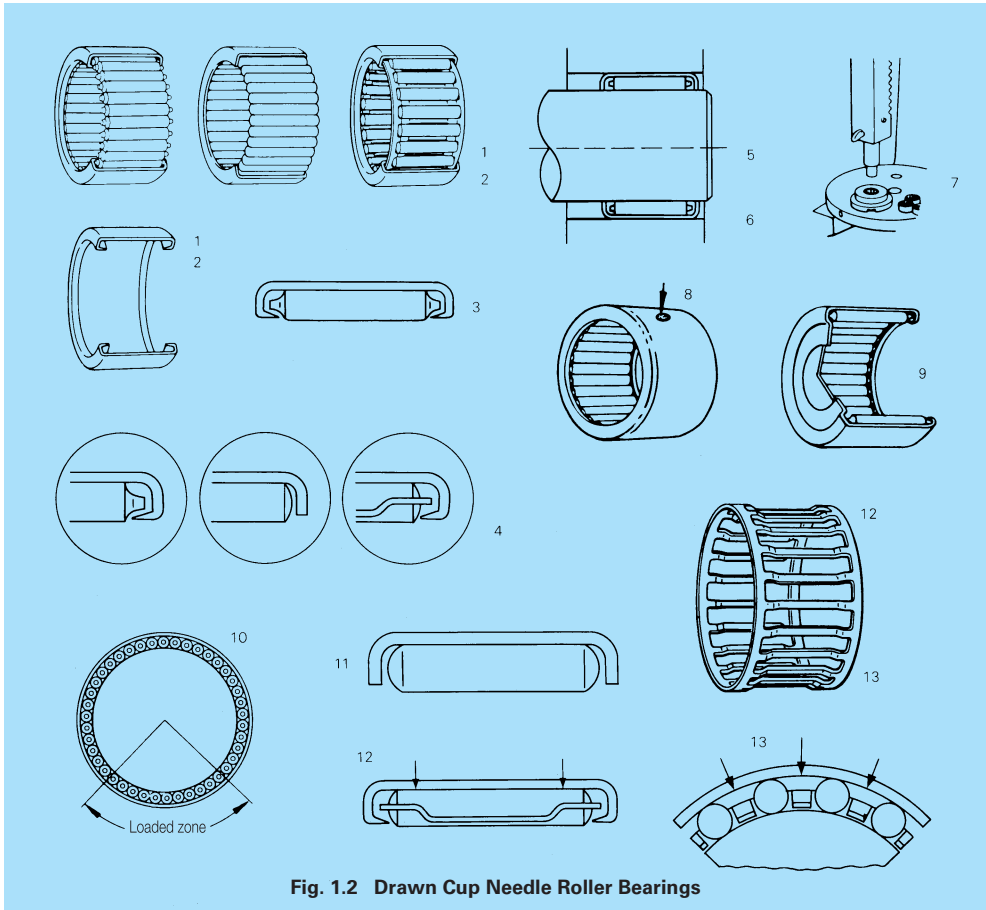


Fig. 1.2 Drawn Cup Needle Roller Bearings

1.3 Solid Needle Roller Bearings

1. The bearing ring can sustain high shock loads because it is precisely ground after being heat treated.
2. Providing high load capacity, many rollers are held in the outer ring by strong integrated ribs.
3. The unique single-piece cage is also excellent for high speeds as the cage accurately guides the rollers at the roller pitch diameter at both ends of the rollers.
4. Considerable space is allotted for lubricant and lubricant life is long.
5. The cage is surface treated, increasing wear resistance and rigidity.
6. Appropriate crowning is applied to the rollers to avoid edge load.
7. Outer rings provided with a lubrication groove and lubrication hole facilitate the replenishment of lubricant.
8. As the size variation per unit of roller diameter is small, the distribution of the roller load is favorable and bearing life is long.

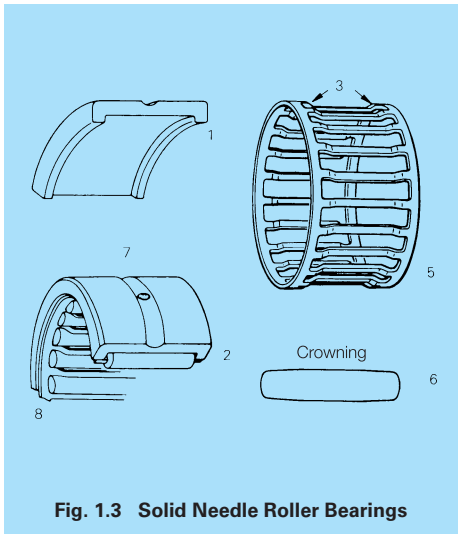


Fig. 1.3 Solid Needle Roller Bearings

1.4 Thrust Bearings

1.4.1 Thrust Needle Bearings

1. Making compact design feasible, the height of the cross-section is similar to that of a conventional thrust washer.
2. The cage, which is made of two steel sheets precisely pressed, holds and accurately guides the rollers. The cage is surface hardened to increase wear resistance and rigidity.
3. There are thrust washers of various thicknesses available to meet the requirements of different applications.

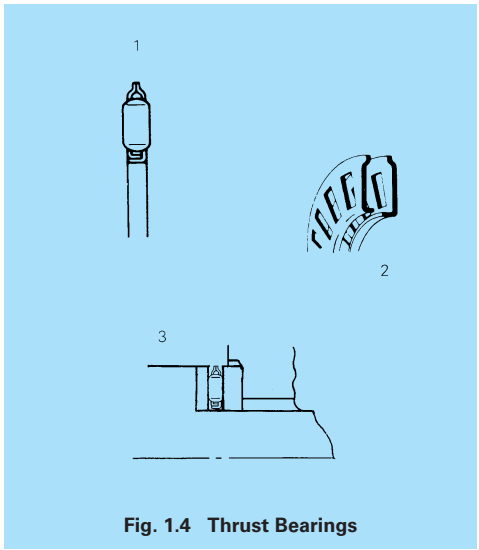


Fig. 1.4 Thrust Bearings

1.5 Cam Followers and Roller Followers

1. The thick-walled outer ring made of high-carbon chromium bearing steel, which is finished by grinding after hardening, has high shock resistance and is highly resistant to deformation.
2. The cam follower has lubrication holes in three places. Lubrication can be provided easily from any of these three locations.
3. Crowning is applied to the rollers in order to avoid edge loading.
4. The labyrinth seal structure reduces friction. In addition, a special thrust washer is integrated in order to enhance heat seizure resistance.
5. Increasing shock resistance, the shaft of the cam follower has a hardened raceway surface.
6. The roller follower can carry heavy loads because both ends are supported.

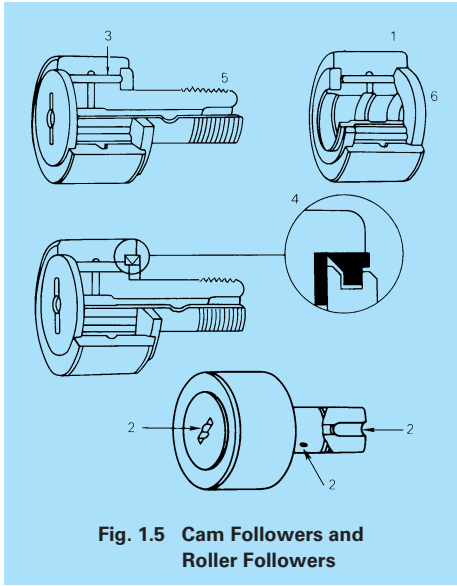


Fig. 1.5 Cam Followers and Roller Followers

1.6 Needle Rollers

There are a variety of end faces available from which an appropriate shape can be selected for specific applications.

1. Rounded rollers are the most widely used.
2. Flat rollers have a longer effective load-carrying length.
3. Crank pin rollers are suitable for where the corner of the shaft or housing step is considerably rounded.
4. Appropriate crowning can be applied for various usage conditions.

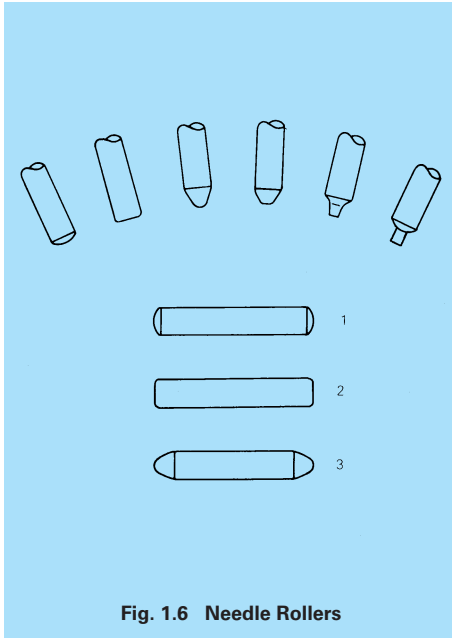


Fig. 1.6 Needle Rollers

1.7 Spherical Plain Bearings

1. Since the structure is composed simply of an inner ring with spherical outside and an outer ring with a spherical bore, this bearing is ideal for applications where the inner and outer rings are misaligned.
2. Carefully selected high-carbon chromium bearing steel is used, and high load capacity is achieved thanks to ideal hardening.
3. Bearings with single-cut outer rings are not separable even without snap rings.
4. Bearings with double-cut outer rings are assembled by a wire snap ring into a single body.
5. The lubrication grooves and lubrication holes provided in the inner and outer rings facilitate re-lubrication.
6. After phosphate coating, a coating of molybdenum disulfide is applied to the whole surface.

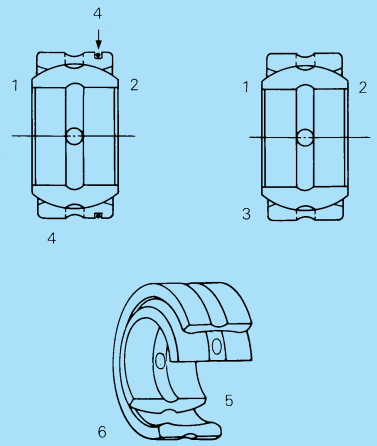


Fig. 1.7 Spherical Plain Bearings

1.8 Needle Roller Bearings for Universal Joints

Needle roller bearings for universal joints include two types: drawn cup and solid.

1. While the tightening method is mainly used for securing drawn cup needle bearings in a yoke hole, another method employs a rose washer and bearing assembly is carried out using special equipment. Universal joints made by assembling drawn cup needle bearings have the following characteristics:
   (a) It is possible to impose a light preload between the bearing inner bottom and cross shaft end.
   (b) It is possible to have the cross shaft's rotating center aligned with the shaft center.
   (c) Universal joints made in this way are economical and contribute to making products more compact and lightweight.
2. As solid needle bearings can sustain heavier loads than drawn cup needle bearings, universal joints with solid needle bearings are suitable for medium and heavy-duty motor vehicles.

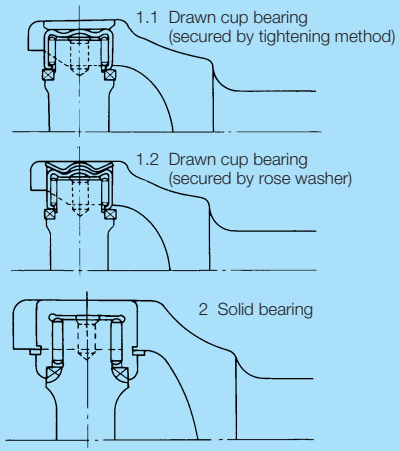


Fig. 1.8 Needle Roller Bearings for Universal Joints

1.9 Drawn Cup Roller Clutches

There are three types with a clutch only: FC, FCL and RC; and three types with a clutch and bearing combined: FCB, RCB and FCBN.

1. The unique structure of the drawn cup outer ring makes the clutch lightweight and compact.
2. The height of the cross-section is the same as standard drawn cup needle bearings and the torque capacity is high.
3. Performance is accurate and friction torque is low during overrunning.
4. Installation can be done in the same way as for drawn cup needle bearings, that is, by simply press fitting the unit into the housing.

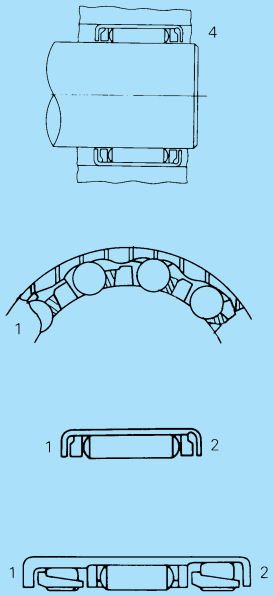


Fig. 1.9 Drawn Cup Roller Clutches

2. Bearing Life and Basic Load Rating

2.1 Bearing Life

Like other rolling bearings, needle roller bearings cannot be used forever and can fracture for various reasons. If bearings are incorrectly mounted, lubrication is improper or bearings are exposed to dust and moisture due to poor sealing, they will eventually fail to perform satisfactorily. Even if bearings are properly mounted and operated correctly with sufficient lubrication, when rolling bearings are operated under load, the inner and outer ring raceways and rolling elements are subjected to repeated cyclic stress. Because of metal fatigue of the rolling contact surfaces of the raceways and rolling elements, scaly particles may separate from the bearing material. This phenomenon is called "flaking". Bearing fatigue life is defined by the total number of revolutions after which the bearing surface will start flaking due to stress. If the speed is constant, the fatigue life is often expressed by the total number of operating hours. As shown in **Fig. 2.1**, even for seemingly identical bearings, which are of the same type, size and material, and receive the same heat treatment and other processing, the rolling fatigue life varies greatly even under identical operating conditions. This is because the flaking of materials due to fatigue is subject to many other variables. Consequently, "rating fatigue life" is used in preference to actual rolling fatigue life. Suppose a number of bearings of the same type are operated individually under the same conditions. After a certain period of time, 10% of them fail as a result of flaking caused by rolling fatigue. The total number of revolutions at this point is defined as the basic rating life or, if the speed is constant, the basic rating life is often expressed by the total number of operating hours completed.

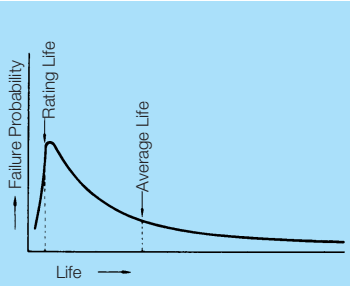


Fig. 2.1
Failure Probability and Bearing Life

2.2 Basic Dynamic Load Rating

The basic load rating is defined as the constant load applied on bearings with stationary outer rings that the inner rings or shafts can endure for a rating life of one million revolutions (10^6 rev). The basic load rating of radial bearings is defined as a central radial load of constant direction and magnitude, while the basic load rating of thrust bearings is defined as an axial load of constant magnitude in the same direction as the central axis. The basic load ratings of roller bearings are defined in ISO 281 and can be calculated by Equation **2.1** and **2.2**. The basic load ratings C_r and C_a listed in the dimension tables of NSK needle bearings are calculated in accordance with ISO, but amended with development of bearing materials and manufacturing quality.

$$C_r = b_m f_c (iL_{we}\cos\alpha)^{\frac{7}{9}} Z^{\frac{3}{4}} D_w^{\frac{29}{27}} \quad \text{.....(2.1)}$$

For thrust roller bearings

$$C_a = b_m f_{ca} L_{we}^{\frac{7}{9}} Z^{\frac{3}{4}} D_w^{\frac{29}{27}} \quad \text{.....(2.2)}$$

where, C_r, C_a : Basic dynamic load rating (N) {kgf}
i : Number of roller rows per bearing
α : Nominal contact angle
Z : Number of rolling elements per row
D_w : Roller diameter
L_{we} : Effective length of roller
b_m : Rating factor depending on normal material and manufacture quality
f_c, f_{ca} : Coefficients determined by contact surface shape between roller and bearing ring, processing accuracy and bearing material

2.3 Life Equation

The following relation exists between the basic dynamic load rating, the bearing load and the rating fatigue life of roller bearings:

$$L = \left(\frac{C_r}{P}\right)^{\frac{10}{3}} \text{ or } L = \left(\frac{C_a}{P}\right)^{\frac{10}{3}} \quad \text{.....(2.3)}$$

where, L : Rating fatigue life (10^6-revolution unit)
P : Bearing load (equivalent load) (N), {kgf}
C_r, C_a : Basic dynamic load rating of radial bearing and thrust bearing (N), {kgf}

When the bearing is rotating at a certain speed n (min^{-1}), it is convenient to express the fatigue life of the bearing in hours. When fatigue life is expressed in hours, 500 hours is referred to as standard,

$$33.3\text{min}^{-1} \times 60\text{min} \times 500\text{h} = 10^6\text{rev}$$

In other words, the basic dynamic load rating C, which gives a life of 10^6 revolutions, gives the life of 500 hours at 33.3min^{-1}

L_h : Fatigue life hours of bearing (h)
....As shown in **Fig. 2.2** and **Appendix 13**
f_h : Fatigue life factor
....As shown in **Fig. 2.2** and **Appendix 13**
f_n : Speed factor
....As shown in **Fig. 2.2** and **Appendix 12**

With the above, the relation between equations **2.4**, **2.5** and **2.6** is obtained.

$$L_h = 500\, f_h^{\frac{10}{3}} \quad \text{.....(2.4)}$$

$$f_h = f_n \cdot \frac{C_r}{P} \text{ or } f_h = f_n \cdot \frac{C_a}{P} \quad \text{.....(2.5)}$$

$$f_n = \left(\frac{33.3}{n}\right)^{\frac{3}{10}} \quad \text{.....(2.6)}$$

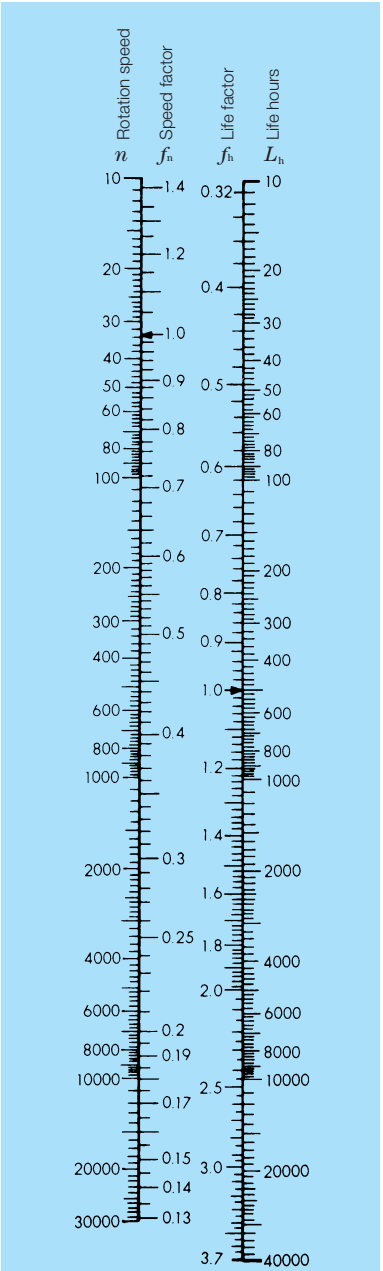


Fig. 2.2 Speed Factor and Life Factor

2.3.1 Temperature Adjustment for Basic Dynamic Load Rating

If rolling bearings are used at high temperature, the hardness of the bearing steel decreases. Consequently, the basic load rating, which depends on the physical properties of the material, also decreases. Therefore, the basic load rating should be adjusted for higher temperatures using the following equation.

$$C_t = f_t \cdot C \quad \text{.....(2.7)}$$

where, C_t : Basic load rating after temperature correction (N), {kgf}
f_t : Temperature factor (**See Table 2.1**)
C : Basic load rating before temperature adjustment (N), {kgf}

Table 2.1 Temperature Factor, f_t

Bearing Temperature ℃	125	150	175	200	250
Temperature Factor f_t	1.00	1.00	0.95	0.90	0.75

2.3.2 Hardness Factor, f_H

The raceway surface should be HRC58 to 64 and hardened to the proper depth. Methods for hardening the raceway surface include full hardening, carburized hardening or induction hardening. A decrease in hardness results in a rapid reduction in the fatigue life of the bearing. When hardness is lower than HRC58, it is necessary to correct the basic dynamic load rating by multiplying it by the hardness factor, f_H, shown in **Fig. 2.3**.

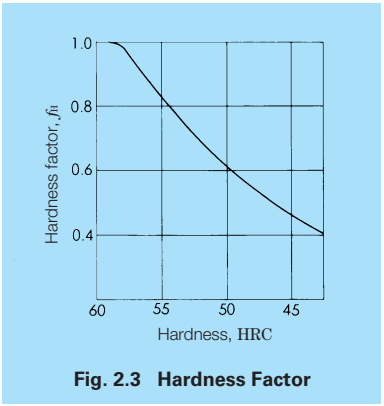


Fig. 2.3 Hardness Factor

2.3.3 Correction of Basic Rating Life

As described previously, the equation for calculating the basic rating life of roller bearings is:

$$L_{10} = \left(\frac{C}{P}\right)^{\frac{10}{3}} \quad \text{.....(2.8)}$$

The L_{10} Life is defined as the basic rating life with a statistical reliability of 90%. Depending on the machines in which the bearings are used, sometimes a reliability higher than 90% may be required.

However, recent improvements in bearing material have greatly extended the fatigue life. In addition, the development of the Elasto-Hydrodynamic Theory of Lubrication proves that the thickness of the lubricating film in the contact zone between rings and rolling elements greatly influences bearing life. To reflect such improvements in the calculation of fatigue life, the basic rating life is adjusted using the following adjustment factors:

$$L_{na} = a_1 a_2 a_3 L_{10} \quad \text{.....(2.9)}$$

where, L_{na} : Adjusted rating life in which reliability, material improvements, lubricating condition, etc. are considered
L_{10} : Basic rating life with a reliability of 90%
a_1 : Life adjustment factor for reliability
a_2 : Life adjustment factor for special bearing properties
a_3 : Life adjustment factor for operating conditions

The life adjustment factor for reliability, a_1, is listed in **Table 2.2** for reliabilities higher than 90%.

The life adjustment factor for special bearing properties, a_2, is greater than one because of improvements in bearing steel.

NSK now uses vacuum degassed bearing steel, and the results of tests by NSK show that life is greatly improved when compared with earlier materials. The basic load ratings C_r and C_a listed in the bearing tables were calculated considering the extended life achieved by improvements in materials and manufacturing techniques. Consequently, when estimating life using Equation **2.9**, it is sufficient to assume $a_1 = 1$.

Table 2.2 Reliability Factor, a_1

Reliability (%)	90	95	96	97	98	99
a_1	1.00	0.62	0.53	0.44	0.33	0.21

The life adjustment factor for operating conditions, a_3, is used to adjust for various factors, particularly lubrication. If there is no misalignment between the inner and outer rings and the thickness of the lubricating film in the contact zones of the bearing is sufficient, it is possible for a_3 to be greater than one; however, a_3 is less than one in the following cases:

- When the viscosity of the lubricant in the contact zones between the raceways and rolling elements is low.
- When the circumferential speed of the rolling elements is very slow.
- When the bearing temperature is high.
- When the lubricant is contaminated by water or foreign matter.
- When misalignment of the inner and outer rings is excessive.

It is difficult to determine the proper value for a_3 for specific operating conditions because there are still many unknowns. Since the special bearing property factor a_2 is also influenced by the operating conditions, there is a proposal to combine a_2 and a_3 into one quantity ($a_2 \times a_3$), and not consider them independently. In this case, under normal lubricating and operating conditions, the product ($a_2 \times a_3$) should be assumed equal to one. However, if the viscosity of the lubricant is too low, the value drops to as low as 0.2.

If there is no misalignment and a lubricant with high viscosity is used so sufficient fluid-film thickness is secured, the product of ($a_2 \times a_3$) can be as high as two.

When selecting a bearing based on the basic load rating, it is best to choose a reliability factor (a_1) appropriate for the projected use and an empirically determined C/P or f_h value derived from past results for lubrication, temperature, mounting conditions, etc. in similar machines.

2.4 Static Load Rating

2.4.1 Static Load Ratings

When subjected to an excessive load or a strong shock load, rolling bearings may incur a local permanent deformation of the rolling elements and permanent deformation of the rolling elements raceway surface if the elastic limit is exceeded. The nonelastic deformation increases in area and depth as the load increases, and when the load exceeds a certain limit, the smooth running of the bearing is impeded.

The basic static load rating is defined as that static load which produces the following calculated contact stress at the center of the contact area between the rolling element subjected to the maximum stress and the raceway surface.

For roller bearings 4 000 MPa{408 kgf/mm^2}

In this most heavily stressed contact area, the sum of the permanent deformation of the rolling element and that of the raceway is nearly 0.0001 times the rolling element's diameter. The basic static load rating C_0 is written C_{0r} for radial bearings and C_{0a} for thrust bearings in the bearing tables.

In addition, following the modification of the criteria for basic static load rating by ISO, the new C_0 values for NSK's roller bearings became about 1.5 to 1.9 times the past values. Consequently, the values of the permissible static load factor, f_s, have also changed, so please pay attention to this.

2.4.2 Permissible Static Load Factor

The permissible static equivalent load on bearings varies depending on the basic static load rating and also the application and operating conditions.

The permissible static load factor, f_s, is a safety factor that is applied to the basic static load rating, and it is defined by the ratio in Equation **2.10**. The generally recommended values of f_s are listed in **Table 2.3**. Conforming to the modification of the static load rating, the values of f_s were revised, especially for bearings for which the values of C_0 were increased, please keep this in mind when selecting bearings.

$$f_s = \frac{C_0}{P_0} \quad \cdots\cdots (\mathbf{2.10})$$

where, C_0 : Basic static load rating (N), {kgf}
P_0 : Static equivalent load (N), {kgf}

Table 2.3 Values of Permissible Static Load Factor, f_s

Operating Conditions	Lower Limit of f_s
Low-noise applications	3
Bearings subjected to vibration and shock loads	2
Standard operating conditions	1.5

2.5 Maximum Permissible Load of Drawn Cup Needle Roller Bearings P_{max}

The drawn cup (outer ring) of a needle bearing is made of a thin-wall special steel sheet that is hardened by carburizing after press-forming. Dynamic loads beyond the maximum permissible load, P_{max}, in the Bearing Tables should not be imposed. For static loads, the permissible load is up to 1.3 times P_{max}.

3. Bearing Loads

3.1 Equivalent Load

While in some cases the loads applied to bearings are purely radial or axial loads, in most cases the loads are a combination of both. In addition, loads usually fluctuate in both magnitude and direction. For this reason, the loads actually applied to bearings cannot be used for bearing life calculations. A hypothetical load that has a constant magnitude and passes through the center of the bearing (and will give the same bearing life that the bearing would attain under actual conditions of load and rotation) should therefore be estimated. Such a hypothetical load is called the equivalent load.

In the case of radial needle bearings, as the only load that can be applied is radial, the dynamic equivalent load can be obtained from the following equation:

$$P = F_r \quad \cdots\cdots (\mathbf{3.1})$$

where, P : Dynamic equivalent load (N), {kgf}
F_r : Radial load imposed on bearing (N), {kgf}

3.2 Calculation of Bearing Loads

The loads applied to bearings generally include the weight of the body to be supported by the bearings, the weight of the revolving elements themselves, the transmission power of gears and belting and the load produced by the operation of the machine in which the bearings are used. These loads can be theoretically calculated, but some of them are difficult to estimate. Therefore, it becomes necessary to correct the estimates using empirically derived data.

3.2.1 Load Factor

When a radial or axial load has been mathematically calculated, the actual load on the bearing may be greater than the calculated load because of vibration and shock present during operation of the machine. Actual loads may be calculated using the following equations:

$$\left.\begin{aligned} F_r &= f_w \cdot F_{rc} \\ F_a &= f_w \cdot F_{ac} \end{aligned}\right\} \quad \cdots\cdots (\mathbf{3.2})$$

where, F_r, F_a : Loads applied on bearing (N), {kgf}
F_{rc}, F_{ac} : Theoretically calculated load (N), {kgf}
f_w : Load factor

The values given in **Table 3.1** are usually used for the load factor, f_w

Table 3.1 Values of Load Factor, f_w

Operating Conditions	Typical Applications	f_w
Smooth operation free from shock	Electric motors, Machine tools, Air conditioners	1 to 1.2
Normal operation	Air blowers, Compressors, Elevators, Cranes, Papermaking machines	1.2 to 1.5
Operation accompanied by shock and vibration	Construction equipment, Crushers, Vibrating screens, Rolling mills	1.5 to 3

3.2.2 Bearing Loads in Belt or Chain Transmission Applications

The force acting on the pulley or sprocket wheel when power is transmitted by a belt or chain is calculated using the following equations:

$$\left.\begin{aligned} M &= 9\,550\,000 H/n \quad \cdots\cdots (\mathrm{N \cdot mm}) \\ &= 974\,000 H/n \quad \cdots\cdots \{\mathrm{kgf \cdot mm}\} \end{aligned}\right\} \quad \cdots\cdots (\mathbf{3.3})$$

$$P_k = M/r \quad \cdots\cdots (\mathbf{3.4})$$

where, M : Torque acting on pulley or sprocket wheel (N•mm), {kgf•mm}
P_k : Effective force transmitted by belt or chain (N), {kgf}
H : Power transmitted (kW)
n : Speed ($\mathrm{min^{-1}}$)
r : Effective radius of pulley or sprocket wheel (mm)

When calculating the load on a pulley shaft, the belt tension must be included. Thus, to

calculate the actual load, K_b, in the case of a belt transmission, the effective transmitting power is multiplied by the belt factor, f_b, which represents the belt tension. Values of f_b for different types of belts are shown in **Table 3.2**.

$$K_b = f_b \cdot P_k \quad \text{.....(3.5)}$$

In the case of a chain transmission, the values corresponding to f_b should be 1.25 to 1.5.

Table 3.2 Belt Factor, f_b

Type of Belt	f_b
Toothed belts	1.3 to 2
V belts	2 to 2.5
Flat belts with tension pulleys	2.5 to 3
Flat belts	4 to 5

3.2.3 Bearing Loads in Gear Transmission Applications

The loads imposed on gears in gear transmissions vary according to the type of gear. In the simplest case of spur gears, the load is calculated as follows:

$$\left.\begin{aligned} M &= 9\,550\,000H/n \text{.....(N·mm)} \\ &= 974\,000H/n \text{.....\{kgf·mm\}} \end{aligned}\right\} \text{.....(3.6)}$$

$$P_k = M/r \quad \text{.....(3.7)}$$

$$S_k = P_k \tan\theta \quad \text{.....(3.8)}$$

$$K_c = \sqrt{P_k^2 + S_k^2} = P_k \sec\theta \quad \text{.....(3.9)}$$

where, M : Torque applied to gear (N·mm), {kgf·mm}
P_k : Tangential force on gear (N), {kgf}
S_k : Radial force on gear (N), {kgf}
K_c : Combined force imposed on gear (N), {kgf}
H : Power transmitted (kW)
n : Speed (min^{-1})
r : Pitch circle radius of drive gear (mm)
θ : Pressure angle

In addition to the theoretical load calculated above, vibration and shock (which depend on how accurately the gear is finished) should be included by multiplying the theoretically calculated load by the gear factor, f_g.

The values of f_g should generally be those in **Table 3.3**. When vibration from other sources accompanies gear operation, the actual load is obtained by multiplying the load factor by this gear factor.

Table 3.3 Values of Gear Factor, f_g

Gear Finish Accuracy	f_g
Precision ground gears	1 to 1.1
Ordinary machined gears	1.1 to 1.3

3.3 Average Load

3.3.1 Average of Fluctuating Load

When the load applied to bearings fluctuates, an average load that will yield the same bearing life as the fluctuating load should be calculated.

(1) When the relation between load and rotating speed is divided into the following steps (**Fig. 3.1**):

Load F_1 : Speed n_1; Operating time t_1
Load F_2 : Speed n_2; Operating time t_2
⋮ ⋮ ⋮
Load F_n : Speed n_n; Operating time t_n

Then, the average load, F_m, may be calculated using the following equation:

$$F_m = \sqrt[\frac{10}{3}]{\frac{F_1^{\frac{10}{3}} n_1 t_1 + F_2^{\frac{10}{3}} n_2 t_2 + \cdots + F_n^{\frac{10}{3}} n_n t_n}{n_1 t_1 + n_2 t_2 + \cdots\cdots + n_n t_n}} \quad \text{.....(3.10)}$$

where, F_m : Average fluctuating load (N), {kgf}

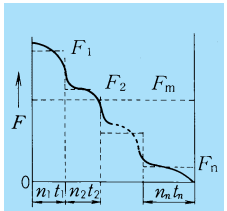


Fig. 3.1 Incremental Load Variation

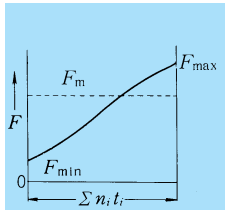


Fig. 3.2 Simple Load Fluctuation

(2) When the load fluctuates almost linearly (**Fig. 3.2**), an approximate value for the average load may be calculated as follows:

$$F_m \fallingdotseq \frac{1}{3}(F_{min} + 2F_{max}) \quad \text{.....(3.11)}$$

where, F_{min}: Minimum value of fluctuating load (N), {kgf}
F_{max}: Maximum value of fluctuating load (N), {kgf}

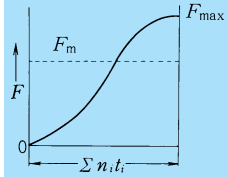


Fig. 3.3

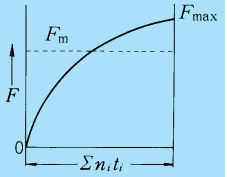


Fig. 3.4

Sinusoidal Load Variation

(3) When the load fluctuation is similar to a sine wave, an approximate value for the average load may be calculated from the following equation:

In the case of **Fig. 3.3**

$$F_m \fallingdotseq 0.65F_{max} \quad \text{.....(3.12)}$$

In the case of **Fig. 3.4**

$$F_m \fallingdotseq 0.75F_{max} \quad \text{.....(3.13)}$$

4. Bearing Selection

4.1 Bearing Types

It is important to select the needle roller bearing that is most suited to the particular application. One must fully understand the structure and characteristics of a selected bearing to determine whether it is appropriate for a given application.

When selecting needle bearings, the following factors should be considered:

(1) Amount, direction and condition of bearing load
(2) Rotating speed and which ring is rotating
(3) Required bearing life
(4) Ambient temperature of bearing
(5) Accuracy of bearings
(6) Degree of friction
(7) Noise
(8) Vertical shaft or horizontal shaft
(9) Lubricating method and sealing device
(10) Mounting and dismounting
(11) Space for mounting
(12) Finishing accuracy, rigidity, and materials of shaft and housing
(13) Cost requirements

Weighing these factors along with the characteristics of specific bearings, the best bearing for a particular application can be selected.

The characteristics of various NSK needle roller bearings are compared in **Table 4.1**.

4.1.1 Comparison of Bearings with Cages and Full Complement Bearings

Comparing bearings of the same size, full complement bearings, both at rest and when rotating, have a higher load rating and maximum permissible load than bearings with cages, and are therefore preferred for most applications.

As the rollers are accurately guided in bearings with a cage and no friction occurs between the rollers, the friction torque is lower than in the full complement roller bearings.

Under ideal lubrication conditions and normal load, approximate friction torque can be obtained with the following equation.

Table 4.1 Comparison of Characteristics of Needle Roller Bearings

Bearing Type / Characteristics	Cage & Needle Rollers	Drawn Cup		Solid	Thrust	Needle Roller	Cam Follower/ Roller Follower	
		With Cage	Full Complement	With Cage	Needle		With Cage	Full Complement
	FWJ, WJ, FBN, etc.	FJ, FJL, J, JH, etc.	F, B, BH, etc.	RNA, HJ, RLM, etc.	FNTA, NTA	——	FCJ, FYCJ	FCR, CR, FYCR, YCR
Radial Load Capacity	High	Medium	High	High	None	Very high	Medium	High
Axial Load Capacity	None	None	None	None	High	None	None	None
Maximum Permissible Load	High	Low	Low	High	High	Very high	Low	Low
Limiting Speed	Very high	Very high	Medium	Very high	High	Medium	High	Low
Permissible Shaft Deflection and Mounting Error	Medium	Medium	Low	Medium	Low	Low	Medium	Low
Grease Life	Very high	Very high	Medium	Very high	Low	Medium	High	Low
Friction Torque	Very low	Very low	Low	Very low	Low	Low	Very low	Low
Cross-Section Height	Very low	Very low	Very low	Medium	Very low	Very low	Medium	Medium
Cost performance	High	High	High	Moderate	High	Very high	Moderate	Moderate

Friction torque of a full complement roller bearing:

$$0.0025 \times \text{pitch circle radius of rollers} \times \text{bearing load}$$

Friction torque of cage assembly:

$$0.0015 \times \text{pitch circle radius of rollers} \times \text{bearing load}$$

While full complement roller bearings can be operated at considerably high speeds (see the limiting speeds shown in the Bearing Tables), bearings with cages are recommended for high-speed applications. Bearings with cages are also preferable for applications where shaft deflection or mounting error is relatively high because their rollers are accurately guided and crowned.

4.1.2 Comparison between Drawn Cup and Solid Bearings

Employing a thick, hardened bearing ring, solid bearings can sustain high shock loads and continuous loads. Drawn cup bearings are not to be used beyond the maximum permissible load shown in the Bearing Tables.

Although solid bearings of high rigidity can be used with double-cut, split housings, drawn cup bearings are affected by bore accuracy and housing rigidity due to their thin outer ring. Therefore, double-cut, split housings should be avoided.

When mounting solid bearings, positioning is required to be in line with the shoulder of the housing hole or a snap ring. However, with drawn cup bearings, as through-holes are sufficient for housings, bore processing is easier and more economical.

Table 4.2 Fatigue Life Factors, f_h, for Various Bearing Applications

Operating Periods	Fatigue Life Factor, f_h				
	≤ 3	2 to 4	3 to 5	4 to 7	≥ 6
Infrequently used or only for short periods	· Small motors for home appliances like vacuum cleaners and washing machines · Hand power tools	· Agricultural equipment			
Used only occasionally but reliability is important		· Motors for home heaters and air conditioners · Construction equipment	· Conveyors · Elevator cable sheaves		
Used intermittently for relatively long periods	· Rolling mill roll necks	· Small motors · Deck cranes · General cargo cranes · Pinion stands · Passenger cars	· Factory motors · Machine tools · Transmissions · Vibrating screens · Crushers	· Crane sheaves · Compressors · Specialized transmissions	
Used intermittently for more than eight hours daily		· Escalators	· Centrifugal separators · Air conditioning equipment · Blowers · Woodworking machines · Large motors · Axle boxes on railway rolling stock	· Mine hoists · Press flywheels · Railway traction motors · Locomotive axle boxes	· Papermaking machines
Used continuously and high reliability is important					· Waterworks pumps · Electric power stations · Mine draining pumps

4.2 Machinery in which Bearings are Used and Projected Life

It is not advisable to select bearings with unnecessarily high load ratings, for such bearings may be too large and uneconomical. In addition, the bearing life alone should not be the deciding factor in the selection of bearings. The strength, rigidity, and design of the shaft on which the bearings are to be mounted should also be considered. Bearings are used in a wide range of applications and the design life varies with specific applications and operating conditions. **Table 4.2** gives empirical fatigue life factors derived from operating experience for various applications.

4.3 Limiting Speed

Rolling bearings are subject to speed limitations. When bearings are operating, the higher the speed, the higher the bearing temperature due to friction. The limiting speed is the empirically obtained value for the maximum speed at which bearings can be continuously operated without failing from seizure or generation of excessive heat. Naturally, limiting speeds (in min^{-1}) vary depending on such factors as bearing type and dimensions, cage type and material, bearing load, lubricating method, and cooling conditions, including those of the bearing's surroundings. When bearing load, P, exceeds 8% of the basic load rating, C, the limiting speed shown in the Bearing Tables must be corrected by multiplying it by the correction factor in **Fig. 4.1**.

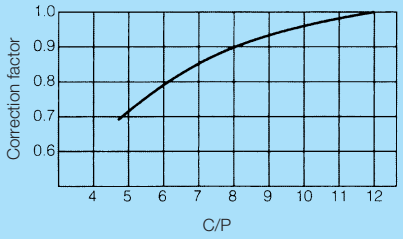


Fig. 4.1
Correction of Limiting Speed by Bearing Load

In the Bearing Tables, the limiting speeds for oil lubrication are shown for each bearing. For grease lubrication, consider 60 to 70% of the oil lubrication values as a guide. Please consult NSK for further details.

4.4 Internal Clearance of Rolling Bearings

The internal clearance of rolling bearings during operation is one of the key features affecting bearing characteristics such as life span, heat generation, vibration and noise. Internal clearance is the space between the raceway ring and the rolling elements. The distance the free ring moves when a specific measured radial load is applied is defined as the radial measured clearance. In roller bearings, since elastic deformation caused by the measured load can be ignored, the measured clearance and geometric clearance (radial internal clearance) are considered to be the same.

Values for the radial internal clearance of metric solid needle bearings with interchangeable components are listed in **Table 4.3**. These are classified from smallest clearance to largest into C2, CN, C3, C4 and C5. When the clearance range of bearings with interchangeable rings is too wide, matched bearings with the clearances listed in **Table 4.4** (page A23) are available.

The radial internal clearances of inch needle bearings with inner rings, HJ + IR, are shown in **Table 4.5** (page A25). HJ + IR needle bearings

Table 4.3 Interchangeable Radial Internal Clearances of Metric Solid Needle Roller Bearings

unit : μm

Bearing Bore Diameter d (mm)		Clearance									
		C2		CN		C3		C4		C5	
over	incl	min	max	min	max	min	max	min	max	min	max
–	10	0	25	20	45	35	60	50	75	–	–
10	24	0	25	20	45	35	60	50	75	65	90
24	30	0	25	20	45	35	60	50	75	70	95
30	40	5	30	25	50	45	70	60	85	80	105
40	50	5	35	30	60	50	80	70	100	95	125
50	65	10	40	40	70	60	90	80	110	110	140
65	80	10	45	40	75	65	100	90	125	130	165
80	100	15	50	50	85	75	110	105	140	155	190
100	120	15	55	50	90	85	125	125	165	180	220
120	140	15	60	60	105	100	145	145	190	200	245
140	160	20	70	70	120	115	165	165	215	225	275
160	180	25	75	75	125	120	170	170	220	250	300
180	200	35	90	90	145	140	195	195	250	275	330
200	225	45	105	105	165	160	220	220	280	305	365
225	250	45	110	110	175	170	235	235	300	330	395
250	280	55	125	125	195	190	260	260	330	370	440
280	315	55	130	130	205	200	275	275	350	410	485
315	355	65	145	145	225	225	305	305	385	455	535
355	400	100	190	190	280	280	370	370	460	510	600
400	450	110	210	210	310	310	410	410	510	565	665

Table 4.4 Matched Radial Internal Clearance of Metric Solid Needle Roller Bearings

unit : μm

Bearing Bore Diameter d (mm)		Clearance									
		CC2		CC		CC3		CC4		CC5	
over	incl	min	max	min	max	min	max	min	max	min	max
6	10	10	20	20	30	35	45	45	55	–	–
10	14	10	20	20	30	35	45	45	55	65	75
14	18	10	20	20	30	35	45	45	55	65	75
18	24	10	20	20	30	35	45	45	55	65	75
24	30	10	25	25	35	40	50	50	60	70	80
30	40	12	25	25	40	45	55	55	70	80	95
40	50	15	30	30	45	50	65	65	80	95	110
50	65	15	35	35	50	55	75	75	90	110	130
65	80	20	40	40	60	70	90	90	110	130	150
80	100	25	45	45	70	80	105	105	125	155	180
100	120	25	50	50	80	95	125	120	145	180	205
120	140	30	60	60	90	105	135	135	160	200	230
140	160	35	65	65	100	115	150	150	180	225	260
160	180	35	75	75	110	125	165	165	200	250	285
180	200	40	80	80	120	140	180	180	220	275	315
200	225	45	90	90	135	155	200	200	240	305	350
225	250	50	100	100	150	170	215	215	265	330	380
250	280	55	110	110	165	185	240	240	295	370	420
280	315	60	120	120	180	205	265	265	325	410	470
315	355	65	135	135	200	225	295	295	360	455	520
355	400	75	150	150	225	255	330	330	405	510	585
400	450	85	170	170	255	285	370	370	455	565	650

are wide and clearance values are determined broadly based on previous experience to be on the safe side. As the values in **Table 4.3** and **4.4** are also used for cylindrical roller bearings, which are narrower, CN clearance is not commonly used for needle bearings, which are wider and produced in various structures.

Clearance at a certain temperature and under particular rotation conditions, even under conditions where the load causes no elastic deformation, is defined as the effective clearance. Theoretically, bearing life is longest when the effective clearance is slightly negative. It is, however, difficult to operate all bearings under these ideal conditions and if the proper negative clearance is exceeded, bearing life significantly decreases. The effective clearance, therefore, should generally be set as positive. Fitting conditions, temperature conditions and mounting error should be taken into consideration in clearance selection.

4.4.1 Fitting Condition

Radial clearance is reduced by expansion or contraction of the bearing rings when fitting either an inner or an outer ring onto a shaft or into a housing with interference. This decreased value, obtained by the calculation shown in **Table 5.1** (page A31), is approximately 75 to 90% of interference.

As the fitting surface is finished with a certain tolerance, it is advisable that calculation of the decreased value in internal clearance due to fitting be dealt with statistically.

4.4.2 Bearings and Circumference Temperature Conditions

Frictional heat generated by bearing rotation is released through shafts and housings. Since heat is released more readily through the housing, the temperature of the inner ring and rolling elements is higher than that of the outer ring. This difference in temperature is larger when the shaft is heated or the housing is cooled. Radial internal clearance is reduced by the difference in heat expansion resulting from the temperature difference between the inner and outer rings. The value of this clearance reduction in rolling bearings is calculated by the following equation:

$$\delta_t \fallingdotseq \frac{1}{4}\alpha \Delta_t (3D+d) \quad \text{.....(4.1)}$$

where, δ_t: Decreased value of radial internal clearance by temperature difference between inner and outer ring (mm)
α : Coefficient of linear expansion of bearing steel (12.5×10^{-6})
Δ_t: Temperature difference between inner and outer rings (°C)
D: Nominal outside diameter (mm)
d: Nominal bore diameter (mm)

4.4.3 Shaft Deflection and Mounting Error of Bearings

When shaft deflection and mounting error are large, it is necessary to select a larger bearing clearance. For metric solid needle bearings, for example, when the fitting of the inner and outer rings is tight (i.e., tighter than the values of k and K) care should be taken to select a clearance of C3 or CC3 and larger from **Tables 4.3** and **4.4**, in order to prevent insufficient radial internal clearance.

For solid needle bearings, which are often used without an inner ring, various radial internal clearances are obtained by selecting shaft tolerances, as shown in **Table 4.6**.

Table 4.5 Interchangeable Radial Internal Clearance of Inch Solid Needle Roller Bearings

unit: μm

Nominal Dimension of Roller's Inscribed Circle Diameter, F_w mm (inch)		Clearance	
over	incl	min	max
——	15.875 (5/8)	33	66
15.875 (5/8)	25.400 (1)	41	77
25.400 (1)	28.575 (1 1/8)	46	82
28.575 (1 1/8)	34.925 (1 3/8)	48	86
34.925 (1 3/8)	41.275 (1 5/8)	50	88
41.275 (1 5/8)	47.625 (1 7/8)	50	91
47.625 (1 7/8)	69.850 (2 3/4)	56	99
69.850 (2 3/4)	76.200 (3)	56	104
76.200 (3)	95.250 (3 3/4)	63	117
95.250 (3 3/4)	101.600 (4)	68	122
101.600 (4)	114.300 (4 1/2)	68	127
114.300 (4 1/2)	139.700 (5 1/2)	71	132
139.700 (5 1/2)	152.400 (6)	76	137
152.400 (6)	165.100 (6 1/2)	76	142
165.100 (6 1/2)	196.850 (7 3/4)	84	154
196.850 (7 3/4)	209.550 (8 1/4)	84	159
209.550 (8 1/4)	234.950 (9 1/4)	87	162

Note: For internal clearances other than the values shown in this table, please consult NSK.

Table 4.6 Fitting Tolerances and Radial Internal Clearance of Shafts Assembled with Solid Needle Roller Bearings without Inner Rings

Nominal Dimension of Roller's Inscribed Circle Diameter, F_w (mm)		C2	CN	C3	C4
over	incl				
6	180	k5	g5	f6	e6
180	315	j6	f6	e6	d6
315	490	h6	e6	d6	c6

Note: For tight fitting of the outer ring (class K and higher) select a smaller shaft dimension in consideration of the contraction of the inscribed circle diameter after assembly.

4.4.4 Radial Internal Clearance of Drawn Cup Needle Roller Bearings

Because the correct form and dimensional accuracy of drawn cup needle bearings is achieved only after press fitting into a specific housing bore, the method for selecting the internal clearance of solid needle bearings cannot be applied to drawn cup bearings. Please refer to page B39 for the internal clearances of drawn cup needle bearings.

4.5 Bearing Accuracy

NSK needle roller bearings are manufactured in conformity with the accuracy classes set down by ISO. Class 0 is adequate for general use, while bearings with higher accuracy, such as Classes 6, 5, and 4, are recommended for use when extremely low runout is required, when the rotating speed is high, and/or when less friction and shaft fluctuation are required. As various factors, such as bearing type, internal clearance, lubrication, and mounting, should be considered when selecting higher-accuracy bearings, please consult NSK.

As defined by ISO, the boundary dimensions for radial bearings, tolerances for boundary dimensions and running accuracy, and limits for chamfer dimensions are shown in **Table 4.7**. Regarding high-accuracy bearings of Class 6 and higher, please consult NSK.

Tolerances for Radial Bearings and Widths of Outer Rings

Table 4.7.1 Tolerances for Inner Rings

unit: μm

Nominal Bore Diameter d (mm) over	incl	Δ_{dmp}(2) Normal high	Normal low	Class 6 high	Class 6 low	Class 5 high	Class 5 low	Class 4 high	Class 4 low	Δ_{ds}(2) Class 4 Diameter Series 0,1,2,3,4 high	Δ_{ds}(2) Class 4 Diameter Series 0,1,2,3,4 low	V_{dp}(2) Normal Diameter Series 9 max	Normal 0,1 max	Normal 2,3,4 max	Class 6 Diameter Series 9 max	Class 6 0,1 max	Class 6 2,3,4 max	Class 5 Diameter Series 9 max	Class 5 0,1,2,3,4 max	Class 4 Diameter Series 9 max	Class 4 0,1,2,3,4 max
0.6 (1)	2.5	0	−8	0	−7	0	−5	0	−4	0	−4	10	8	6	9	7	5	5	4	4	3
2.5	10	0	−8	0	−7	0	−5	0	−4	0	−4	10	8	6	9	7	5	5	4	4	3
10	18	0	−8	0	−7	0	−5	0	−4	0	−4	10	8	6	9	7	5	5	4	4	3
18	30	0	−10	0	−8	0	−6	0	−5	0	−5	13	10	8	10	8	6	6	5	5	4
30	50	0	−12	0	−10	0	−8	0	−6	0	−6	15	12	9	13	10	8	8	6	6	5
50	80	0	−15	0	−12	0	−9	0	−7	0	−7	19	19	11	15	15	9	9	7	7	5
80	120	0	−20	0	−15	0	−10	0	−8	0	−8	25	25	15	19	19	11	10	8	8	6
120	150	0	−25	0	−18	0	−13	0	−10	0	−10	31	31	19	23	23	14	13	10	10	8
150	180	0	−25	0	−18	0	−13	0	−10	0	−10	31	31	19	23	23	14	13	10	10	8
180	250	0	−30	0	−22	0	−15	0	−12	0	−12	38	38	23	28	28	17	15	12	12	9
250	315	0	−35	0	−25	0	−18	–	–	–	–	44	44	26	31	31	19	18	14	–	–
315	400	0	−40	0	−30	0	−23	–	–	–	–	50	50	30	38	38	23	23	18	–	–
400	500	0	−45	0	−35	–	–	–	–	–	–	56	56	34	44	44	26	–	–	–	–

Notes (1) 0.6 mm is included in the group.
(2) Applicable to bearings with cylindrical bores.
(3) Tolerance for width deviation and tolerance limits for the width variation of the outer ring should be the same for a given bearing. Tolerances for the width variation of the outer ring of Class 5, 4, and 2 are shown in **Table 4.7.2**.

Remarks The cylindrical bore diameter "no-go side" tolerance limits (high) specified in this table do not necessarily apply within a distance of 1.2 times the chamfer dimension r (max) from the ring face.

Table 4.7.2 Tolerances for Outer Rings

unit: μm

Nominal Outside Diameter D (mm) over	incl	Δ_{Dmp} Normal high	Normal low	Class 6 high	Class 6 low	Class 5 high	Class 5 low	Class 4 high	Class 4 low	Δ_{Ds} Class 4 Diameter Series 0,1,2,3,4 high	Δ_{Ds} Class 4 Diameter Series 0,1,2,3,4 low	V_{Dp} Normal Open Type Diameter Series 7,8,9 max	Normal 0,1 max	Normal 2,3,4 max	Class 6 Open Type Diameter Series 7,8,9 max	Class 6 0,1 max	Class 6 2,3,4 max	Class 5 Open Type Diameter Series 7,8,9 max	Class 5 0,1,2,3,4 max	Class 4 Open Type Diameter Series 7,8,9 max	Class 4 0,1,2,3,4 max
2.5 (1)	6	0	−8	0	−7	0	−5	0	−4	0	−4	10	8	6	9	7	5	5	4	4	3
6	18	0	−8	0	−7	0	−5	0	−4	0	−4	10	8	6	9	7	5	5	4	4	3
18	30	0	−9	0	−8	0	−6	0	−5	0	−5	12	9	7	10	8	6	6	5	5	4
30	50	0	−11	0	−9	0	−7	0	−6	0	−6	14	11	8	11	9	7	7	5	6	5
50	80	0	−13	0	−11	0	−9	0	−7	0	−7	16	13	10	14	11	8	9	7	7	5
80	120	0	−15	0	−13	0	−10	0	−8	0	−8	19	19	11	16	16	10	10	8	8	6
120	150	0	−18	0	−15	0	−11	0	−9	0	−9	23	23	14	19	19	11	11	8	9	7
150	180	0	−25	0	−18	0	−13	0	−10	0	−10	31	31	19	23	23	14	13	10	10	8
180	250	0	−30	0	−20	0	−15	0	−11	0	−11	38	38	23	25	25	15	15	11	11	8
250	315	0	−35	0	−25	0	−18	0	−13	0	−13	44	44	26	31	31	19	18	14	13	10
315	400	0	−40	0	−28	0	−20	0	−15	0	−15	50	50	30	35	35	21	20	15	15	11
400	500	0	−45	0	−33	0	−23	–	–	–	–	56	56	34	41	41	25	23	17	–	–
500	630	0	−50	0	−38	0	−28	–	–	–	–	63	63	38	48	48	29	28	21	–	–

Notes (1) 2.5 mm is included in the group.
(2) The tolerances for outer ring width variation of bearings of Classes Normal and 6 are shown in **Table 4.7.1**.

Remarks The outside diameter "no-go side" tolerances (low) specified in this table do not necessarily apply within a distance of 1.2 times the chamfer dimension r (max) from the ring face.

Table 4.7.3 Tolerances for Radial Bearings and Widths of Outer Rings

unit: μm

Δ_{dmp}(2) Normal max	Class 6 max	Class 5 max	Class 4 max	Δ_{Bs}(or Δ_{Cs})(3) Single Bearing Normal Class 6 high	Normal Class 6 low	Class 5 Class 4 high	Class 5 Class 4 low	V_{Bs}(or V_{Cs}) Inner Ring (or Outer Ring) (3) Normal max	Inner Ring (or Outer Ring) (3) Class 6 max	Inner Ring Class 5 max	Inner Ring Class 4 max	K_{ia} Normal max	Class 6 max	Class 5 max	Class 4 max	S_d Class 5 max	Class 4 max
6	5	3	2	0	−40	0	−40	12	12	5	2.5	10	5	4	2.5	7	3
6	5	3	2	0	−120	0	−40	15	15	5	2.5	10	6	4	2.5	7	3
6	5	3	2	0	−120	0	−80	20	20	5	2.5	10	7	4	2.5	7	3
8	6	3	2.5	0	−120	0	−120	20	20	5	2.5	13	8	4	3	8	4
9	8	4	3	0	−120	0	−120	20	20	5	3	15	10	5	4	8	4
11	9	5	3.5	0	−150	0	−150	25	25	6	4	20	10	5	4	8	5
15	11	5	4	0	−200	0	−200	25	25	7	4	25	13	6	5	9	5
19	14	7	5	0	−250	0	−250	30	30	8	5	30	18	8	6	10	6
19	14	7	5	0	−250	0	−250	30	30	8	5	30	18	8	6	10	6
23	17	8	6	0	−300	0	−300	30	30	10	6	40	20	10	8	11	7
26	19	9	–	0	−350	0	−350	35	35	13	–	50	25	13	–	13	–
30	23	12	–	0	−400	0	−400	40	40	15	–	60	30	15	–	15	–
34	26	–	–	0	−450	0	–	50	45	–	–	65	35	–	–	–	–

Table 4.7.4 Tolerances for Outer Rings

unit: μm

V_{Dmp} Normal max	Class 6 max	Class 5 max	Class 4 max	K_{ea} Normal max	Class 6 max	Class 5 max	Class 4 max	S_D Class 5 max	Class 4 max	V_{Cs}(2) Class 5 max	Class 4 max
6	5	3	2	15	8	5	3	8	4	5	2.5
6	5	3	2	15	8	5	3	8	4	5	2.5
7	6	3	2.5	15	9	6	4	8	4	5	2.5
8	7	4	3	20	10	7	5	8	4	5	2.5
10	8	5	3.5	25	13	8	5	8	4	6	3
11	10	5	4	35	18	10	6	9	5	8	4
14	11	6	5	40	20	11	7	10	5	8	5
19	14	7	5	45	23	13	8	10	5	8	5
23	15	8	6	50	25	15	10	11	7	10	7
26	19	9	7	60	30	18	11	13	8	11	7
30	21	10	8	70	35	20	13	13	10	13	8
34	25	12	–	80	40	23	–	15	–	15	–
38	29	14	–	100	50	25	–	18	–	18	–

Symbols for Boundary Dimensions and Running Accuracy

d Nominal bearing bore diameter
Δ_{ds} Deviation of a single bore diameter
Δ_{dmp} Single plane mean bore diameter deviation
V_{dp} Bore diameter variation in a single radial plane
V_{dmp} Mean bore diameter variation

Δ_{Bs} Deviation of a single inner ring width
V_{Bs} Inner ring width variation

K_{ia} Radial runout of assembled bearing inner ring
S_d Inner ring reference face (backface, where applicable) runout with bore

D Nominal bearing outside diameter
Δ_{Ds} Deviation of a single outside diameter
Δ_{Dmp} Single plane mean outside diameter deviation
V_{Dp} Outside diameter variation in a single radial plane
V_{Dmp} Mean outside diameter variation

Δ_{Cs} Deviation of a single outer ring width
V_{Cs} Outer ring width variation

K_{ea} Radial runout of assembled bearing outer ring
S_D Variation of bearing outside surface generatrix inclination with outer ring reference face (backface)

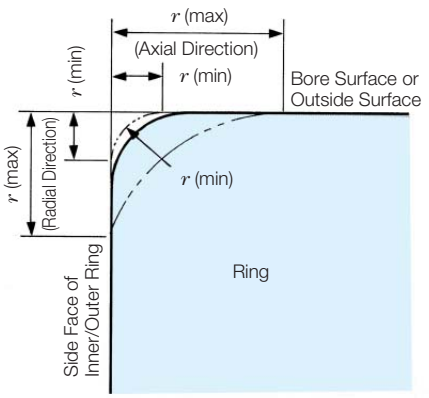
Chamfer Dimension Limits (for Metric Design Bearings)

Table 4.8.1 Chamfer Dimension Limits for Radial Bearings

Units: mm

Permissible Chamfer Dimension for Inner/Outer Rings r (min)	Nominal Bore Diameter d over	incl	Permissible Chamfer Dimension for Inner/Outer Rings r (max) Radial Direction	Axial Direction
0.15	—	—	0.3	0.6
0.2	—	—	0.5	0.8
0.3	—	40	0.6	1
	40	—	0.8	1
0.5	—	—	1.2	1.7
0.6	—	40	1	2
	40	—	1.3	2
1	—	50	1.5	3
	50	—	1.9	3
1.1	—	120	2	3.5
	120	—	2.5	4
1.5	—	120	2.3	4
	120	—	3	5
2	—	80	3	4.5
	80	220	3.5	5
	220	—	3.8	6
2.1	—	280	4	6.5
	280	—	4.5	7
2.5	—	100	3.8	6
	100	280	4.5	6
	280	—	5	7
3	—	280	5	8
	280	—	5.5	8
4	—	—	6.5	9

Remarks For bearings with nominal widths less than 2 mm, the value of r (max) in the axial direction is the same as that in the radial direction.



r : Chamfer Dimension of Inner/Outer Ring

Remarks The precise shape of chamfer surfaces has not been specified but its profile in the axial plane shall not intersect an arc of radius r (min) touching the side face of an inner ring and bore surface, or the side face of an outer ring and outside surface.

5. Shaft and Housing Design

The function of the bearing cannot be achieved fully without adequate design of the bearing abutment, even though appropriate bearings can be selected from a wide range of bearing types. In particular, the raceway ring of needle bearings is very thin compared with other roller bearings, so special consideration should be taken regarding shaft and housing design and fitting.

5.1 Fitting

5.1.1 Purpose of Fitting

The purpose of fitting is to fix an inner ring or outer ring into the shaft or housing and to prevent harmful circumferential slipping, which is called "creep." Creep between fitting surfaces causes wear and results in abrasive metallic particles entering the bearing, which leads to abnormal temperature rise and vibration. Fitting is also necessary for obtaining the correct form and dimensional accuracy for drawn cup needle bearings.

5.1.2 Type and Volume of Load

It is necessary to know which ring, inner or outer, rotates against the load. In the case of an inner ring rotating load, creep occurs when the fitting between the inner ring and shaft is not tight enough. Because creep cannot be prevented simply by fixing the inner ring axially, the fitting for the inner ring and shaft needs to be tight. However, relatively loose fitting is often employed to make mounting and dismounting easier. In this case, the fitting surfaces must be well lubricated to prevent fretting and scoring.

In the case of an outer ring rotating load, the fitting between the outer ring and housing should be tight. For loads of indeterminate direction, such as unbalanced loads, vibration loads, and irregular dimensional loads, tight fitting of both the inner and outer rings is recommended.

In accordance with the size of the load, the inner ring is compressed in the radial direction and becomes relatively wider while interference decreases. The interference must not, however, be allowed to disappear due to the radial load.

The decrease in interference, Δd_F, is estimated using the following equation:

$$\left.\begin{aligned}\Delta d_F &= 0.08\sqrt{\frac{d}{B}}F_r\times10^{-3} \quad \cdots\cdots(\mathrm{N})\\ \Delta d_F &= 0.25\sqrt{\frac{d}{B}}F_r\times10^{-3} \quad \cdots\cdots\{\mathrm{kgf}\}\end{aligned}\right\} \cdots\cdots(\mathbf{5.1})$$

where, Δd_F: Decrease in interference of the inner ring (mm)
d: Nominal bore diameter (mm)
B: Nominal inner ring width (mm)
F_r: Radial load applied on bearing (N), {kgf}

5.1.3 Effective Interference and Finish of Shaft and Housing

Since the roughness of fitted surfaces is reduced during fitting, the effective interference becomes less than the apparent interference. The amount of this interference decrease varies depending on the roughness of the surfaces and may be estimated using the following equations:

For ground shafts $\quad \Delta d \fallingdotseq \dfrac{d}{d+2}\Delta d_a\,(\mathrm{mm})$ .....(**5.2**)

For machined shafts $\quad \Delta d \fallingdotseq \dfrac{d}{d+3}\Delta d_a\,(\mathrm{mm})$ .....(**5.3**)

where, Δd: Effective interference (mm)
Δd_a: Apparent interference (mm)
d : Bearing nominal bore diameter (mm)

According to Equations **5.2** and **5.3**, the effective interference of bearings with a bore diameter of 30 to 150 mm is about 95% of the apparent interference.

5.1.4 Interference Variation Caused by Temperature Difference between Bearing and Shaft or Housing

The effective interference decreases due to the increasing bearing temperature during operation. If the temperature difference between the bearing and housing is ΔT (°C), then the temperature difference between the fitted surfaces of the shaft and inner ring is estimated to be about (0.1 to 0.15) ΔT in case that the shaft is cooled. The decrease in the interference of the inner ring due to this temperature difference Δd_T may be calculated using Equation **5.4**:

$$\begin{aligned}\Delta d_T &= (0.10 \text{ to } 0.15)\,\Delta T\cdot\alpha\cdot d\\ &\fallingdotseq 0.0015\,\Delta T\cdot d\times10^{-3}\,(\mathrm{mm})\end{aligned} \cdots\cdots(\mathbf{5.4})$$

where, Δd_T: Decrease in interference of inner ring due to temperature difference (mm)
ΔT: Temperature difference between bearing interior and surrounding parts (°C)
α: Coefficient of linear expansion of bearing steel $= 12.5\times10^{-6}$ (1/°C)
d: Nominal bearing bore diameter (mm)

In addition, depending on the temperature difference between the outer ring and housing, or difference in their coefficients of linear expansion, the interference may increase.

5.1.5 Deformation and Stress Caused by Fitting

When mounting a bearing with an interference fit, deformation and stress of the rings result. Calculations of surface pressure, stress, and expansion and contraction of inner and outer rings during fitting are the same as those of a thick cylinder to which is applied internal or external pressure. These calculations are compiled and shown in **Table 5.1**. The modulus of longitudinal elasticity and Poisson's ratio of shaft and housing are calculated as the inner and outer rings are the

Table 5.1 Fit Calculations

	Inner ring and shaft	Outer ring and housing
Surface pressure p_m (MPa) (kgf/mm²)	Hollow shaft $p_m=\dfrac{\Delta d}{d}\dfrac{1}{\left[\dfrac{m_s-1}{m_sE_s}-\dfrac{m_i-1}{m_iE_i}\right]+2\left[\dfrac{k_0^2}{E_s(1-k_0^2)}+\dfrac{1}{E_i(1-k^2)}\right]}$ Solid shaft $p_m=\dfrac{\Delta d}{d}\dfrac{1}{\left[\dfrac{m_s-1}{m_sE_s}-\dfrac{m_i-1}{m_iE_i}\right]+\left[\dfrac{2}{E_i(1-k^2)}\right]}$	Housing outside diameter $p_m=\dfrac{\Delta D}{D}\dfrac{1}{\left[\dfrac{m_e-1}{m_eE_e}-\dfrac{m_h-1}{m_hE_h}\right]+2\left[\dfrac{h^2}{E_e(1-h^2)}+\dfrac{1}{E_h(1-h_0^2)}\right]}$
Expansion of inner ring raceway ΔD_i (mm) Contraction of outer ring raceway ΔD_e (mm)	$\Delta D_i = 2d\dfrac{p_m}{E_i}\dfrac{k}{1-k^2}$ $= \Delta d\cdot k\dfrac{1-k_0^2}{1-k^2k_0^2}$ (hollow shaft) $= \Delta d\cdot k$ (solid shaft)	$\Delta D_e = 2D\dfrac{p_m}{E_e}\dfrac{1}{1-h^2}$ $= \Delta D\cdot h\dfrac{1-h_0^2}{1-h^2h_0^2}$
Maximum stress $\sigma_{t\,max}$ (MPa) (kgf/mm²)	Circumferential stress at inner ring bore fitting surface is maximum. $\sigma_{t\,max} = p_m\dfrac{1+k^2}{1-k^2}$	Circumferential stress at outer ring bore surface is maximum. $\sigma_{t\,max} = p_m\dfrac{2}{1-h^2}$
Symbols	d : Shaft diameter, inner ring bore d_0 : Hollow shaft bore D_i : Inner ring raceway diameter $k = d/D_i$, $k_0 = d_0/d$ E_i : Inner ring Young's modulus, 208 000 MPa {21 200 kgf/mm²} E_s : Shaft Young's modulus m_i: Inner ring poisson's number, 3.33 m_s: Shaft poisson's number	D : Housing bore diameter, outer ring outside diameter D_0 : Housing outside diameter D_e : Outer ring raceway diameter $h = D_e/D$, $h_0 = D/D_0$ E_e : Outer ring Young's modulus 208 000 MPa {21 200 kgf/mm²} E_h : Housing Young's modulus m_e: Outer ring poisson's number, 3.33 m_h: Housing poisson's number

same. Circumferential stress at the inner ring bore surface should be equal to or less than 127 N/mm², {13 kgf/mm²}.

Table 5.1 does not apply to drawn cup needle bearings.

5.1.6 Recommended Fitting

As discussed above, selecting appropriate fitting requires knowledge of various conditions such as the actual volume and type of load applied to the bearing, temperature conditions, and other conditions affected by fitting. However, it is very difficult to obtain accurate data for all factors involved. Fitting, therefore, is often determined based on previous experience and records. Standard fittings for needle bearings are shown in **Tables 5.2** and **5.3**. Generally, when mounting a bearing in a thin-wall housing or on a hollow shaft, tighter fitting is required.

A split housing cannot be used with drawn cup needle bearings. If there are no alternatives, a bearing press-fitted into a cylindrical sleeve can be used. It is recommended not to use fitting tighter than that of JS7 (J7) for solid bearings in split housings.

The fitting examples shown in **Tables 5.2** and **5.3** (page A33) can be applied to solid shafts and steel or cast iron thick—wall housings. For mounting needle bearings on hollow shafts and in thin-wall or light metal housings, fitting should be tighter than the values shown in **Tables 5.2** and **5.3**. When higher bearing accuracy is especially required, it is necessary to enhance the accuracy of the shafts and housings.

As correct form and dimensional accuracy of the outer ring of drawn cup needle bearings are obtained only after press fitting into a specific housing bore, please refer to page B40 for fittings for drawn cup needle bearings. However, for the fittings for the inner ring and shaft being used for drawn cup needle bearings, refer to the values in **Table 5.2**.

Tolerances for shafts and housing bore diameters are listed in **Appendices 9** and **10** at the end of this volume.

Table 5.2 Fittings between Inner Rings and Shafts of Needle Roller Bearings

Load Conditions		Examples	Shaft Diameter (mm)	Tolerance of Shaft
Rotating Outer Ring Load	Medium-speed rotation with light or normal load	Wheels on stationary axles, rope sheaves, tension pulleys, idle gears	All shaft diameters	g6
	Medium-speed rotation with heavy load			h6
	Accuracy required			h5
Rotating Inner Ring Load or Indeterminate Directional Load	Light loads (less than or equal to 0.06 C_r(1))	Electric appliances, precision machinery, machine tools, pumps, blowers, transport vehicles	≤ 18	h5 or js5 (j5)
			≤ 50	js5 (j5)
			From more than 50 to 150	k6
			≥ 150	m6
	Normal loads (0.06 to 0.13 C_r(1))	General bearings, pumps, main bearings of medium and large engines, woodworking machinery, gears	≤ 50	js5 (j5) or k5
			From more than 50 to 100	m5
			From more than 100 to 150	m6
			From more than 150 to 200	n6
			> 200	p6
	Heavy load (more than 0.13 C_r(1)) or shock loads	Industrial vehicles, construction equipment, crushers	From more than 50 to 150	n6
			> 150	p6

Note (1) C_r represents the basic dynamic load rating of each bearing.

Remarks 1. This table is applicable only to solid steel shafts.
2. For metric bearings, bearings with clearance larger than CN are used when the fitting is greater than k.

Table 5.3 Fittings between Solid Needle Roller Bearings and Housing Bores

Load Conditions		Examples	Tolerance of Bores
Rotating Outer Ring Load	Light or variable loads	Conveyors, rope sheaves, tension pulleys	M7
	Normal or heavy loads	Wheel hubs, crankshafts, connecting rods	N7
	Heavy shock load	Flywheels	P7
Indeterminate Directional Load	Accurate running under light loads	Main spindles of machine tools	K6
	Light or normal loads	Crankshafts, pumps, compressors, large high-speed gears, blowers	JS7 (J7)
	Normal or Heavy Loads		K7
	Shock loads	Eccentric cams	M7
Rotating Inner Ring Load	Accurate running under light or normal loads	Main spindles of machine tools	JS6 (J6)
	Light or normal load	Gears, plummer blocks	H7
	Loads of all kings	General applications	H7 or G7
	Shock loads	Industrial vehicles, construction equipment, crushers	JS7 (J7)

Remarks 1. This table is not applicable to drawn cup needle bearings.
2. This table is applicable to steel and cast iron housings. For light metal housings, fitting should be tighter than in this table.
3. For metric bearings, bearings with clearance larger than CN are used when the fitting is greater than K.

5.2 Accuracy and Roughness

5.2.1 Accuracy and Roughness of Fitting Surfaces

Since the raceway rings of needle bearings are extremely thin, the raceway surface is greatly affected by the accuracy of the shaft and housing. For general operating conditions, a turned finish, smooth bored finish, or reaming finish is acceptable. For high accuracy and low noise under heavy load, however, a grinding finish is required.

A finish roughness of less than1.6μmRa is desirable. As a split housing may deform the outer ring of a thin-wall needle bearing, care should be exercised with the bearing's finish, such as putting a relief on the machining surface of the mating parts. Accuracy and roughness for machining surfaces are shown in **Table 5.4**.

5.2.2 Accuracy and Roughness of the Raceway Surface

The shaft and housing bore are often used as raceway surfaces of needle bearings in order to achieve the most compact bearing design and enhance shaft rigidity, load capacity and accuracy. In this case, accuracy and roughness of the raceway surface greatly affect the life, noise and accuracy of bearings. Therefore, shape, accuracy and roughness have to be treated with great care. In particular, harmful circumferential waviness and a polygonal shape are not desirable. Accuracy and roughness for raceway surfaces are shown in **Table 5.4**. These values may be changed depending on the desired performance.

5.2.3 Shaft Inclination

Shaft inclination caused by external force elastic deflection and mounting error should be less than the values indicated below. If shaft inclination exceeds the values in the table, please consult NSK.

Bearing Width (mm)	Permissible Inclination (mm/mm)	
	With Cage	Full Complement
≤ 25	0.0015	0.0010
(25) to 50	0.0010	0.0005
> 50	0.0005	0.0005

Table 5.4 Shaft and Housing Specifications of Radial Needle Roller Bearings
(Cage & Needle Roller Assemblies/Drawn Cup Bearings/Solid Bearings)

Category	Shaft		Housing Bore	
	Raceway Surface	Fitting Surface	Raceway Surface	Fitting Surface
Out-of-Roundness Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$ to $\frac{IT4}{2}$	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Cylindricity Tolerance	$\frac{IT3}{2}$	$\frac{IT3}{2}$ to $\frac{IT4}{2}$	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Roughness Ra	0.4	0.8	0.8	1.6
Hardness	HRC58 to 64 (HRC60 to 64 is desirable) Appropriate depth of hardening is required.	—	HRC58 to 64 (HRC60 to 64 is desirable) Appropriate depth of hardening is required.	—

Remarks 1. Specifications for shafts and housings of connecting rod cage and roller assemblies are referred to on pages B10 and B11.
2. For standard tolerances (IT), which are generally recommended according to the radius method, please refer to **Appendix 14** (page C66).

5.3 Materials and Heat Treatment of Raceway Surface

The raceway rings and rollers of needle bearings are repeatedly stressed on their relatively small contact surfaces. The materials for raceway rings, rollers and shafts and housings that function as raceways must therefore have high hardness, resistance to permanent deformation and long rolling fatigue life. These materials are also required to be resistant to wear and shock, and have good dimensional stabilization. Common materials used for shafts and housings that function as bearing raceways include:

High-carbon chromium bearing steel (for through hardening)
SUJ2 (JIS G 4805)
Carbon steel for machine construction (for carburizing)
S15CK (JIS G 4051)
Chrome molybdenum steel (for carburizing)
SCM415 to 421 (JIS G 4053)
Chrome steel (for carburizing)
SCr415, 420 (JIS G 4053)
Nickel chrome steel (for carburizing)
SNC415 to 815 (JIS G 4053)
Nickel chrome molybdenum steel (for carburizin)
SNCM220, 415, 420 (JIS G 4053)

Other materials, such as S50C and S55C, can be utilized with through hardening or induction hardening.

The hardened layer, which is tempered at 160 to 180°C after hardening, has to develop a martensite structure that has an even distribution of very fine carbides. In the case of cemented or induction hardening of the raceway surface, the surface hardness should not only be HRC58 to 64 (HRC60 to 64 is preferable), but also the hardened layers with Vickers hardness of HV653 (HRC58) and HV550 (HRC52.3) have to reach appropriate depths. When the values of hardness are below these values, bearing fatigue life significantly decreases (see page A14). The hardened layer depth (up to HV550) after grinding finish is estimated by using the following equation:

$$t \geqq (0.08 \text{ to } 0.10)\, D_w \quad \text{(5.5)}$$

where, t : Effective hardened layer depth (mm)
D_w: Roller diameter (mm)

Core hardness is generally HRC30 to 45.

5.4 Abutment and Fillet Dimensions

The outer ring of solid needle bearings should be positioned in the axial direction using the housing shoulder, collar and snap ring. When mounting, the unmarked side should touch the housing shoulder.

The radius r_a at the rounded fillet of the housing should be less than the minimum value of the bearing's chamfer dimension r. When mounting the inner ring on the shaft, the same consideration should be taken (refer to **Table 5.5**).

Table 5.5 Fillet Radius r_a of Shaft and Housing

Units: mm

Minimum Permissible Chamfer Dimension r	Fillet Radius r_a of Shaft and Housing (max)
0.15	0.15
0.2	0.2
0.3	0.3
0.5	0.5
0.6	0.6
1	1
1.1	1
1.5	1.5
2	2
2.1	2
2.5	2
3	2.5
4	3

The shoulder height h should be less than the width of the side face of the outer ring and less than the thickness of the inner ring. It should also have sufficient surface contact. The shaft and housing shoulder need to be finished with a right angle against the shaft center (**Fig. 5.1**).

The marked side of drawn cup needle bearings should be in contact with the mounting jig, and mounted onto the step-less through hole. Drawn cup needle bearings should not be positioned in the axial direction by either a snap ring or collar. The side face of the drawn cup outer ring has to be 0.25 mm or more shorter than the housing's side face in order to protect the bending part of the outer ring's side face (**Fig. 5.2**).

Other abutment and fillet dimensions are presented in the introductory section for each series. Please refer to the related sections as required.

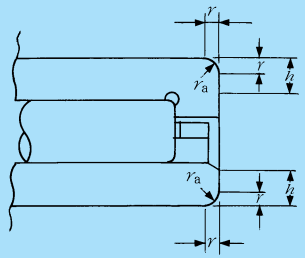


Fig. 5.1

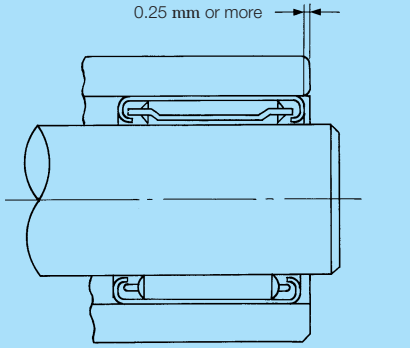


Fig. 5.2

6. Lubrication and Lubricating Methods

6.1 Friction and Lubrication

Low friction in a bearing means low power loss and temperature rise, which together enable machines to achieve higher speeds. Friction in a machine may vary depending on bearing type, structure, dimensions, as well as the finishing accuracy of the rolling and sliding contact surfaces. It is also influenced by operational conditions, such as bearing load, rotating speed and lubrication. With light load and high-speed rotation, lubrication largely affects the starting torque, while with heavy load and low rotating speed, the load's effect dominates. Torque converts itself into thermal energy causing temperature rise in bearings. The degree of temperature rise is determined by the difference between the amount of heat generated and that dissipated. When bearing temperature exceeds 120°C, the characteristics of the bearing's material are changed and the lubricating agent is deteriorated.

6.1.1 Purposes of Lubrication

The main purposes of lubrication are to reduce friction and wear inside bearings that may cause premature failure. The effects of lubrication may be briefly explained as follows:

(1) Reduction of Friction and Wear

In needle bearings, it is necessary to lubricate the area where sliding occurs between the raceway surface and rollers due to roller skew and elastic deformation during operation in order to reduce sliding friction and wear between the cage and rollers, the cage and guide surface of the rotating ring, and the roller end face and the ring's rib face.

(2) Extension of Fatigue Life

The rolling fatigue life of bearings depends greatly upon the viscosity and film thickness between the rolling contact surfaces. A heavy film thickness prolongs the fatigue life, but it is shortened if the viscosity of the oil is too low so the film thickness is insufficient.

(3) Dissipation of Frictional Heat and Cooling

Circulation lubrication may be used to carry away frictional heat or heat transferred from overheating and the oil from deteriorating.

(4) Others

Adequate lubrication also helps to prevent foreign material from entering the bearings and guards against corrosion or rusting.

6.2 Lubricants

6.2.1 Lubricating Grease

Grease is a semi-solid lubricant consisting of a base oil and a thickener. The main types and general properties of grease are shown in **Table 6.1**. It should be remembered that different brands of the same type of grease may have different properties.

(1) Base Oil

Mineral oils or synthetic oils such as silicone or diester oil are mainly used as the base oil for grease. The lubrication properties of grease depend mainly on the characteristics of its base oil. Therefore, the viscosity of the base oil is just as important when selecting grease as when selecting an oil. Usually, greases made with low-viscosity base oils are more suitable for high speeds and low temperatures, while greases made with high-viscosity base oils are more suited for high temperatures and heavy loads. However, as the thickener also influences the lubricating properties of grease, the selection criteria for grease are not the same as for lubricating oil.

(2) Thickener

Thickeners for lubricating grease include several types of metallic soaps, inorganic thickeners such as silica gel and bentonite, and heat-resistant organic thickeners such as polyurea and fluoric compounds. The type of thickener is closely related to the grease dropping point ([1]): generally, grease with a high dropping point also has high-temperature capability. This said, grease does not have a

Table 6.1 Grease Properties

Name (Popular name)	Lithium Grease			Sodium Grease (Fiber Grease)	Calcium Grease (Cup Grease)	Mixed Base Grease
Thickener	Li Soap			Na Soap	Ca Soap	Na+Ca Soap, Li+Ca Soap, etc.
Base Oil	Mineral Oil	Diester Oil, Polyatomic Ester Oil	Silicon Oil	Mineral Oil	Mineral Oil	Mineral Oil
Properties						
Dropping Point, °C	170 to 195	170 to 195	200 to 210	170 to 210	70 to 90	160 to 190
Working Temperatures, °C	−20 to + 110	−50 to + 130	−50 to + 160	−20 to + 130	−20 to + 60	−20 to + 80
Working Speed, % (1)	70	100	60	70	40	70
Mechanical Stability	Good	Good	Good	Good	Poor	Good
Pressure Resistance	Fair	Fair	Poor	Fair	Poor	Fair to Good
Water Resistance	Good	Good	Good	Poor	Good	Poor for Na Soap Grease
Rust Prevention	Good	Good	Poor	Poor to Good	Good	Fair to Good
Remarks	General purpose grease used for numerous applications	Good low temperature and torque characteristics. Often used for small motors and instrument bearings. Pay attention to rust caused by insulation varnish.	Mainly for high-temperature applications. Unsuitable for bearings for high and low speeds or heavy loads or those having numerous sliding-contact areas (roller bearings, etc.)	Long and short fiber types are available. Long fiver grease is unsuitable for high speeds. Attention to water and high temperature is required.	Extreme pressure grease containing high-viscosity mineral oil and extreme pressure additive (Pb soap, etc.) has high pressure resistance.	Often used for roller bearings and large ball bearings.

Complex Grease	Non-Soap Based Grease (Non-Soap Grease)	
Ca Complex Soap, Al Complex Soap, Li Complex Soap, etc.	Urea, Bentonite, Carbon Black, Fluoric Compounds, Heat Resistant Organic Compound, etc.	
Mineral Oil	Mineral Oil	Synthetic Oil (Ester Oil, Polyatomic Ester Oil, Synthetic Hydrocarbon Oil, Silicone Oil, Fluoric Oil)
180 to 300	≥230	≥230
−20 to + 130	−10 to + 130	≤ + 220
70	70	40 to 100
Good	Good	Good
Fair to Good	Fair	Fair
Good	Good	Good
Fair to Good	Fair to Good	Fair to Good
Suitable for extreme pressures mechanically stable	Mineral oil base grease is middle and high temperature purpose lubricant. Synthetic oil base grease is recommended for low or high temperature. Some silicone and fluoric oil-based greases have poor rust prevention and noise performance.	

Note (1) The values listed are percentages of the limiting speeds given in the bearing tables.
Remarks The grease properties shown here can vary between brands.

Table 6.2 Consistency and Working Conditions

Consistency Number	0	1	2	3	4
Consistency (1) 1/10 mm	355 to 385	310 to 340	265 to 295	220 to 250	175 to 205
Working Conditions (Application)	• For centralized oiling • When fretting is likely to occur	• For centralized oiling • When fretting is likely to occur • For low temperature	• For general use	• For general use • For high temperature	• For high temperature • For grease seals

Note (1) Consistency: Depth into grease attained by a cone when pressed with a specified weight, indicated in units of 1/10 mm. The larger the value, the softer the grease.

high working temperature unless its base oil is heat-resistant. The highest possible working temperature for grease should be determined considering the heat resistance of the base oil. The water resistance of grease depends upon the type of thickener. Sodium soap grease or compound grease containing sodium soap emulsifies when exposed to water or high humidity and therefore cannot be used where moisture is prevalent.

Note (1) The grease dropping point is that temperature at which a grease heated in a specified small container becomes sufficiently fluid to drip.

(3) Additives

Grease often contains various additives such as antioxidants, corrosion inhibitors, and extreme pressure additives to give it special properties. It is recommended that extreme pressure additives be used in heavy load applications. For long use without replenishment, an antioxidant should be added.

(4) Consistency

Consistency refers to the "softness" of grease. **Table 6.2** shows the relation between consistency and working conditions.

(5) Mixing Different Types of Grease

In general, different brands of grease must not be mixed. Mixing grease with different types of thickeners may destroy its composition and physical properties. Even if the thickeners are of the same type, possible differences in additives may cause detrimental effects.

6.2.2 Lubricating Oil

Highly refined mineral or synthetic oil is used as lubricant for rolling bearings. The first point to consider when selecting lubricating oil is appropriate viscosity. This varies depending on the type of needle bearings. Under regular operating conditions, lubricating oil with a viscosity of 13 mm^2/s (cSt) or more at the operating temperature should be selected. For thrust bearings, oil with a higher viscosity should be chosen. Generally, higher rotational speeds require lower viscosity and larger loads and larger bearing sizes require higher viscosity.

In addition to proper viscosity, characteristics required of lubricating oil include cleanliness, resistance to corrosion, high lubrication performance, superior oxidation resistance, fluidity at low temperatures and pressure resistance. **Fig. 6.1** shows the temperature-viscosity chart for general oil. The viscosity change at a given operating temperature, as well as the appropriate viscosity can be determined from this curve. The method for selecting lubricating oil under various operating conditions is illustrated in **Table 6.3**.

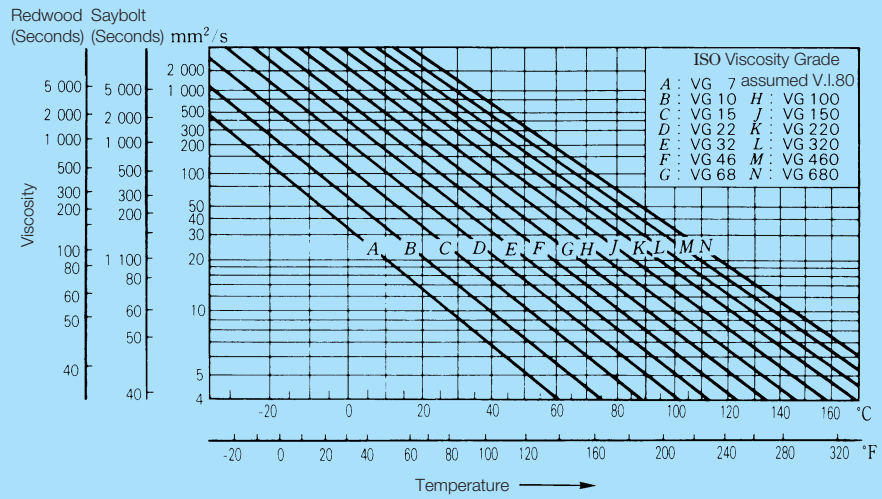


Fig. 6.1 Temperature-Viscosity Chart

Table 6.3 Examples of Selection of Lubricating Oils

Operating Temperature	Speed	Light or Average Load	Heavy or Shock Load
−30 to 0°C	Less than limiting speed	ISO VG 15, 22, 32 (refrigerating machine oil)	—
0 ~ 50°C	Less than 50% of limiting speed	ISO VG 32, 46, 68 (bearing oil, turbine oil)	ISO VG 46, 68, 100 (bearing oil, turbine oil)
	50 to 100% of limiting speed	ISO VG 15, 22, 32 (bearing oil, turbine oil)	ISO VG 22, 32, 46 (bearing oil, turbine oil)
	More than limiting speed	ISO VG 10, 15, 22 (bearing oil)	—
50 ~ 80°C	Less than 50% of limiting speed	ISO VG 100, 150, 220 (bearing oil)	ISO VG 150, 220, 320 (bearing oil)
	50 to 100% of limiting speed	ISO VG 46, 68, 100 (bearing oil, turbine oil)	ISO VG 68, 100, 150 (bearing oil, turbine oil)
	More than limiting speed	ISO VG 32, 46, 68 (bearing oil, turbine oil)	—
80 ~ 110°C	Less than 50% of limiting speed	ISO VG 320, 460 (bearing oil)	ISO VG 460, 680 (bearing oil, gear oil)
	50 to 100% of limiting speed	ISO VG 150, 220 (bearing oil)	ISO VG 220, 320 (bearing oil)
	More than limiting speed	ISO VG 68, 100 (bearing oil, turbine oil)	—

Remarks
1. For the limiting speed, use the values listed in the bearing tables.
2. Refer to Refrigerating Machine Oils (JIS K 2211), Bearing Oils (JIS K 2239), Turbine Oils (JIS K 2213), Gear Oils (JIS K 2219).
3. If the operating temperature is near the high end of the temperature range listed in the left column, select a high viscosity oil.
4. If the operating temperature is lower than −30°C or higher than 110°C, it is advisable to consult NSK.

6.3 Lubricating Methods

6.3.1 Comparison of Grease Lubrication and Oil Lubrication

The lubrication methods for rolling bearings are generally classified into grease lubrication and oil lubrication. Grease lubrication can last from six months to two years before being replenished. It is therefore ideal for reducing maintenance. However, oil lubrication is preferred if the design allows because of its many advantages over grease, including higher limiting speed, longer lubricant life and better cooling properties. **Table 6.4** compares grease and oil lubrication.

Table 6.4 Comparison of Grease and Oil Lubrication

Item	Grease Lubrication	Oil Lubrication
Limiting Speed	Low and normal speed	All speeds
Lubricating Performance	Good	Superior
Cooling Effect	Poor	Good
Lubricant Life	Relatively short	Relatively long
Change of Lubricant	Complicated	Easy
Seal	Easy	Complicated
Lubricant Leakage	Less	Slightly more
Oil Film Strength	Relatively poor	Good
Filtration of Foreign Matter	Difficult	Easy

6.3.2 Grease Lubrication

With grease lubrication, the internal space in the housing is required to be only 1.5 times the bearing's volume. Generally, the bearing is fully packed with grease before mounting and the amount packed into the housing is 1/4 to 1/3 the volume of the free internal space. When the internal space of the housing is less than 1.5 times the bearing's volume, it is appropriate that the filling amount for the housing be 1/2 to 2/3 the volume of the free space including the bearing itself. Grease should be supplied not only to the space inside the housing, but to the inside of the bearing as well.

If too much grease is supplied, churning of the grease will result and the bearing's temperature will rise. This causes deterioration and softening of the grease resulting in leakage. Generally, once grease is packed, refilling isn't necessary for a long period of time. However, the housing should be designed for easy refilling and changing of grease. When frequent refilling of grease is required, such as at high rotating speeds, a grease nipple should be installed and use of a grease gun for refilling is advisable.

6.3.3 Replenishing Interval

Even if high-quality grease is used, there is deterioration of its properties with time so periodic replenishment is required. **Fig. 6.2** shows the replenishment time intervals for various bearing types running at different speeds. This figure applies only if the temperature of the bearing is less than 70°C; the replenishment time interval must be reduced by half for every 15°C temperature rise of the bearings.

The refilling intervals shown in **Fig. 6.2** are for applications where the entry of damaging foreign debris such as dust, dirt and water is limited. Therefore, when such damaging

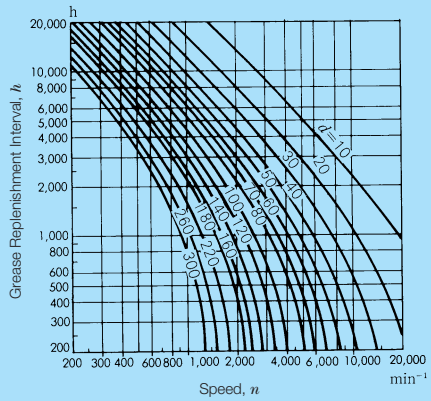


Fig. 6.2 Grease Replenishment Intervals

influences are present, it is necessary to shorten the refilling interval.

6.3.4 Oil Lubrication

(1) Oil Bath Lubrication

Oil bath lubrication is a widely used method in the case of low or medium speeds. The oil level should be at the center of the lowest roller. It is desirable to provide a sight gauge so the proper oil level may be maintained.

(2) Splash Lubrication

With this lubricating method, oil is splashed onto the bearing by a gear or rotating disc installed near the bearing. The entry of oil into the bearing depends on the position of the bearing, bearing type, how the splash is produced, and other factors. Improper lubrication at the beginning of rotation occurs in some cases due to the insufficient supply of oil. The reduction gear with an oil reservoir and device for maintaining the oil level shown in **Fig. 6.3** provides an example of how to prevent such insufficient initial-stage lubrication.

Another effective method of splash lubrication involves an oil reservoir with a V-shaped oil groove and the positioning of the oil drain in a high location. Additionally, installing a device with a magnetic plug to protect against the penetration of wear particles generated by gears is effective.

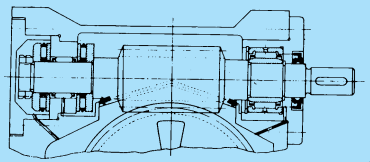


Fig. 6.3 Splash Lubrication

(3) Circulating Lubrication

Circulating lubrication is commonly used for high-speed operation requiring bearing cooling and for bearings used at high temperatures. As shown in **Fig. 6.4**, oil is supplied by the pipe on the right side, it travels through the bearing and drains out through the pipe on the left. After being cooled in a reservoir, it returns to the bearing through a pump and filter. The oil discharge pipe should be larger than the supply pipe so an excessive amount of oil will not back up in the housing.

(4) Drip Feed Lubrication

This method uses an oiler from which a small amount of oil drips into the bearing. The amount of oil supplied varies depending on bearing type. For smaller bearings, the oil supply rate is generally one drop every three to eight seconds. This method can be used at relatively high speeds for normal or light loads.

(5) Other Oil Lubrication Methods

Other methods include jet lubrication and oil mist lubrication. For needle bearings, however, these are not commonly used because of the limitation on limiting speed and space in the vicinity of the bearing.

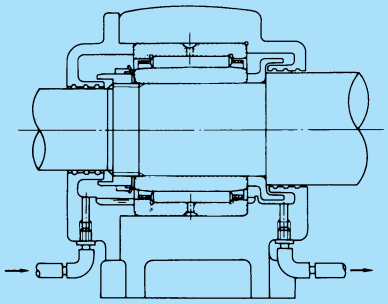


Fig. 6.4 Circulating Lubrication

7. Bearing Seals

To ensure the longest possible life of a bearing, it may be necessary to provide seals to prevent leakage of lubricant and entry of dust, water and other harmful material like metallic particles. The seals must be free from excessive running friction and the probability of seizure. They should also be easy to assemble and disassemble. It is necessary to select a suitable seal for each application considering the lubricating method.

Contact seals are effective because of the contact they maintain with the shaft. They are made of synthetic rubber, synthetic resin, felt or other materials, depending on operating conditions and type of lubricant. Oil seals with synthetic rubber lips are the most common.

7.1 Oil Seals

Synthetic rubber oil seals are usually used in environments where large amounts of dust, dirt, water and other foreign matter can easily enter bearings. Oil seals are standardized in JIS B2402. Suitable oil seals for NSK drawn cup needle roller bearings are listed in the bearing tables on page B242.

When mounting a synthetic rubber oil seal, lubricant should be applied to the seal lip because wear and heat generation easily occur if there is no oil film between the seal lip and shaft. When oil seals are used at high peripheral surface speed or under high internal pressure, the contact surface of the shaft must be smoothly finished and the shaft eccentricity should be 0.02 to 0.05 mm or less.

The contact surface of the shaft should be hardened to enhance its resistance to wear. Generally, hardness of the shaft surface should be higher than HRC40. However, under high-wear conditions, it should be higher than HRC55.

Permissible peripheral speeds and temperature ranges are shown in **Table 7.1**, and criteria for the relationship between shat peripheral surface speed and the roughness of contact surfaces are shown in **Table 7.2**.

Table 7.1 Permissible Peripheral Speeds and Temperature Ranges for Oil Seals

Seal Materials		Permissible Peripheral Speed (m/s)	Operating Temperature Range (℃)(1)
Synthetic Rubber	Nitrile Rubber	≤ 16	−25 to +100
	Acrylic Rubber	≤ 25	−15 to +130
	Silicone Rubber	≤ 32	−70 to +200
	Fluorine Rubber	≤ 32	−30 to +200
Tetrafluoroethylene Resin		≤ 15	−50 to +220

Note (1) The upper limit of the temperature range may be raised by about 20℃ when operating at short intervals.

Table 7.2 Shaft Peripheral Speed and Roughness of Contact Surfaces

Peripheral Speed (m/s)	Surface Roughness R_a
≤ 5	0.8
(5) to 10	0.4
> 10	0.2

7.2 Oil Grooves

Oil grooves, flingers and labyrinths function as non-contact seals. Sealing is achieved by several grooves provided on either or both the shaft and housing. Applying grooves on both surfaces is more efficient for oil lubrication. The clearance between the shaft and housing should be as small as possible. Recommended clearance values are shown in **Table 7.3**.

Table 7.3 Clearances between Shaft and Housing for Oil-Groove Seals

Units: mm

Nominal Shaft Diameter	Radial Clearance
≤ 50	0.25 to 0.4
(50) to 200	0.5 to 1.5

The recommended groove width is approximately 3 to 5 mm, with a depth of about 4 to 5 mm.

When using only grooves for sealing, three or more of them should be used. This sealing method is frequently used to prevent leakage of

oil when the presence of dust and foreign particles is relatively low. Oil grooves are efficient for both oil and grease lubrication at low-speed operation. Filling the groove with grease of a consistency of approximately 200 is most effective.

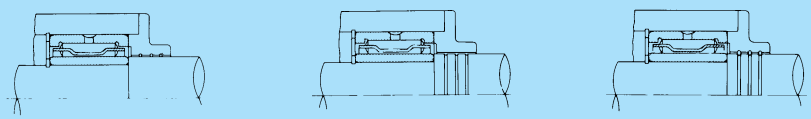


Fig. 7.1 Oil Grooves

7.3 Flingers (Slingers)

Flingers use centrifugal force to throw off oil and dust, and at the same time act as a pump to prevent the leakage of oil and entry of dust.

As shown in **Fig. 7.2**, a flinger used for preventing the entry of dust is effective when combined with other bearing seals. However, due to the fact that the centrifugal force of the flinger pulls the oil inside the housing, care should be exercised during design.

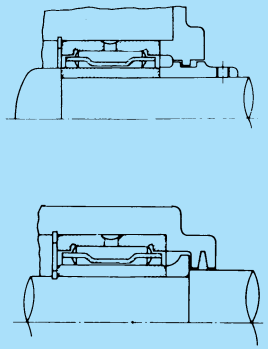


Fig. 7.2 Flingers

7.4 Labyrinth Seals

Among non-contact methods, the labyrinth seal is the most effective and appropriate for preventing oil leakage from a high-speed shaft. The clearance values shown in **Table 7.4** for labyrinth seals are recommended. The sealing effect can be enhanced by filling the clearance with grease. Various structures for labyrinth seals exist. An example is shown in **Fig. 7.3**.

Table 7.4 Clearance for Labyrinth Seals

Units: mm

Nominal Shaft Diameter	Clearance	
	Radial	Axial
≤ 50	0.25 to 0.4	1 to 2
(50) to 200	0.5 to 1.5	2 to 5

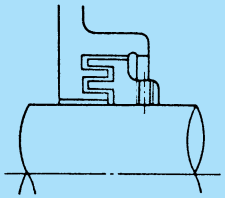


Fig. 7.3 Labyrinth Seal

8. Bearing Handling

8.1 Precautions for Proper Handling of Bearings

Rolling bearings are high-precision machine parts and must be handled accordingly. Even if high-quality bearings are used, their expected performance cannot be achieved if they are not handled properly. The main precautions to be observed are as follows:

(1) Keep Bearings and Surrounding Area Clean

Dust and dirt, even if invisible to the naked eye, have harmful effects on bearings. It is necessary to prevent the entry of dust and dirt by keeping the bearings and their environment as clean as possible.

(2) Handle Bearings Carefully

Heavy shocks during handling may cause bearings to be scratched or otherwise damaged, possibly resulting in their failure. Excessively strong impacts may cause brinelling, breaking, or cracking.

(3) Use Proper Tools

Always use the proper equipment when handling bearings and avoid general-purpose tools.

(4) Prevent Corrosion

As perspiration from hands and various other contaminants may cause corrosion, keep hands clean when handling bearings. Wear gloves if possible. Take care to prevent rusting of bearings caused by corrosive gasses.

8.2 Mounting

Mounting methods vary depending on bearing type and conditions. It is important to fully understand each bearing and handle it properly. In this section, general mounting methods are described. Please note that special precautions apply when mounting drawn cup needle bearings (see page B41).

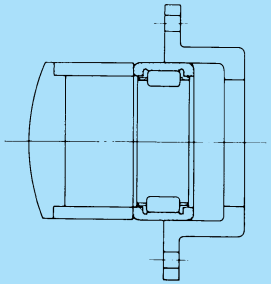


Fig. 8.1
Simultaneous Press Fitting of Outer Rings

(1) Press Fits

Fitting with a press is very common. The mounting dolly is placed onto a raceway ring (an inner or outer bearing ring) that is on the insertion side, then the ring is pushed up into position slowly by a press. An example for an outer ring is shown in **Fig. 8.1**.

(2) Shrink Fits

When mounting an inner ring onto a shaft, shrink fitting is widely used. The maximum heating temperature of the bearing must be no

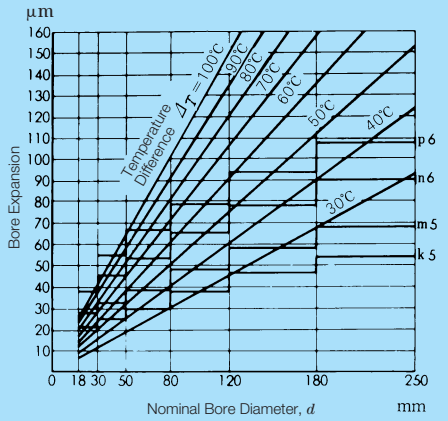


Fig. 8.2
Heat Temperature Differences and Thermal Expansion of Inner Ring